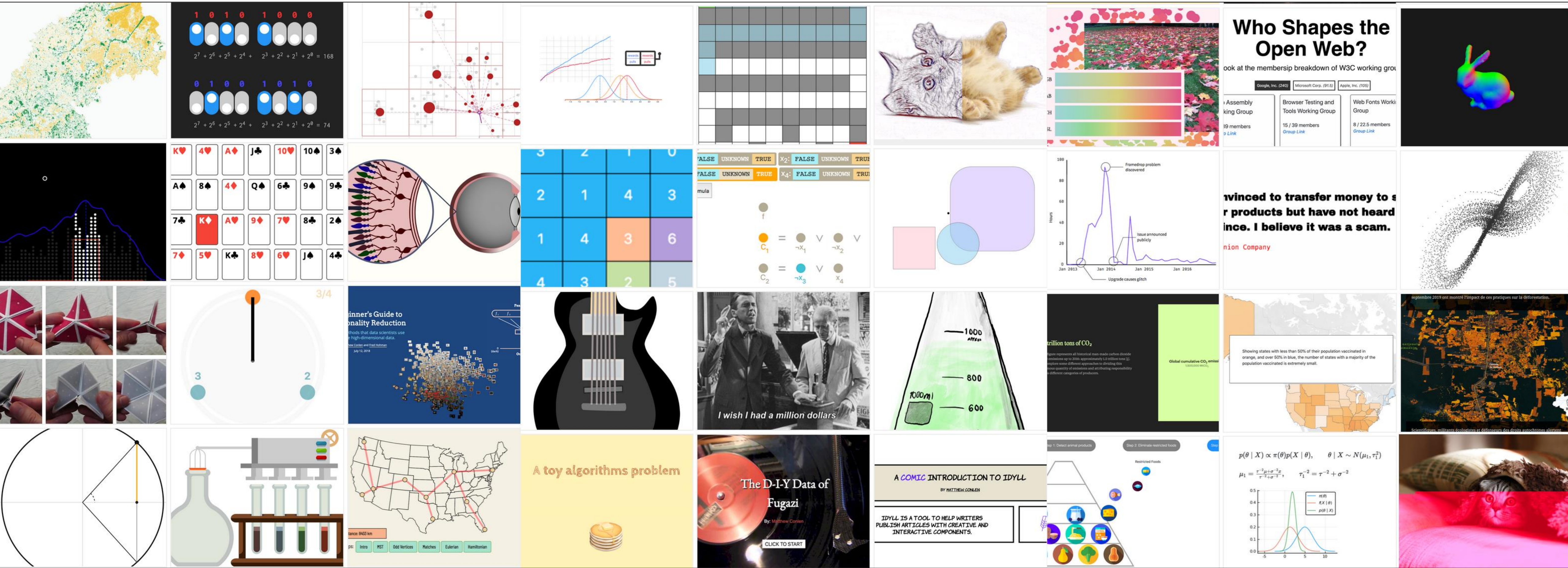


Narrative Visualization



Alex Kale

(slide credited to Matthew Conlen with material from Jeff Heer, Edward Segel)



When Could the United States Reach Herd Immunity? It's Complicated.

By Matthew Conlen and Charlie Smart Feb. 20, 2021

With the [vaccine rollout](#) underway and [coronavirus cases](#) declining after a dark winter surge, it may seem as though the end of the pandemic is in sight. In reality, how soon could we get there?

One answer lies in herd immunity, the point when enough people are immune to the virus that it can no longer spread through the population. Getting there, however, depends not just on how quickly we can vaccinate but on other factors, too, like how many people have already been infected and how easily the virus spreads.

An estimate for the path to herd immunity

100% of population immune



When Could the United States Reach Herd Immunity? It's Complicated.

Conlen & Smart, New York Times 2021

Motivation

Transparency in journalism

“A sense of the tentativeness of truth.”

- Philip Meyer, *Precision Journalism*, 2002

More engaged audience

Greussing & Boomgaarden, *Digital Journalism* 2019

Better learning outcomes

Mayer's Multimedia Principles, 2005

Promote active reading

Victor's "Explorable Explanations", 2011

...also many challenges!

三 Q

The New York Times

[PLAY THE CROSSWORD](#)

[Account](#)

[The Coronavirus Outbreak >](#)

[LIVE Latest Updates](#)

[Maps and Cases](#)

[Vaccine Rollout](#)

[Outdoor Mask Guidance](#)

[Vaccines and Virus Variants](#)

When Could the United States Reach Herd Immunity? It's Complicated.

By Matthew Conlen and Charlie Smart Feb. 20, 2021

With the [vaccine rollout](#) underway and [coronavirus cases](#) declining after a dark winter surge, it may seem as though the end of the pandemic is in sight. In reality, how soon could we get there?

One answer lies in herd immunity, the point when enough people are immune to the virus that it can no longer spread through the population. Getting there, however, depends not just on how quickly we can vaccinate but on other factors, too, like how many people have already been infected and how easily the virus spreads.

An estimate for the path to herd immunity

100% of population immune

This chart shows the current path to herd immunity in the United States, based on a model developed by PHICOR, a public health research group. It looks at the number of people who have been fully vaccinated and combines that with an estimate of the number of people who have been infected and have recovered to measure total immunity.

When the orange line crosses into the blue area, that means we have entered the herd immunity range. The exact threshold for herd immunity for the coronavirus is unknown, but recent estimates range from 70 percent to 90 percent.

At first, this looks like pretty good news — under these assumptions, we could reach herd immunity as early as July. But a lot could happen between now and then. The speed and uptake of vaccination, and how long immunity lasts are big factors. The rise of new virus variants and how we respond to them will also affect the path to herd immunity.

In most scenarios, millions more people will become infected and tens or hundreds of thousands more will die before herd immunity is reached.

What if we speed up vaccinations?

More than 15 million people have been fully vaccinated, and the U.S. is currently administering about 1.7 million shots per day. Some experts say we could nearly double that pace by April as new vaccines are approved. (Because the current vaccines require two doses spaced weeks apart, the number of people fully vaccinated each day is smaller.)

The more people we vaccinate, the faster we could reach the threshold for herd immunity.

CHOOSE A SCENARIO ↕

Topics

Storytelling

- How different mediums can be used to tell stories

Design of narrative visualizations & interactive articles

- Segel & Heer, 2010
- Hohman, et al 2020

Research on Authoring Tools

- Idyll, *UIST 2018*
- Idyll Studio, *UIST 2021*

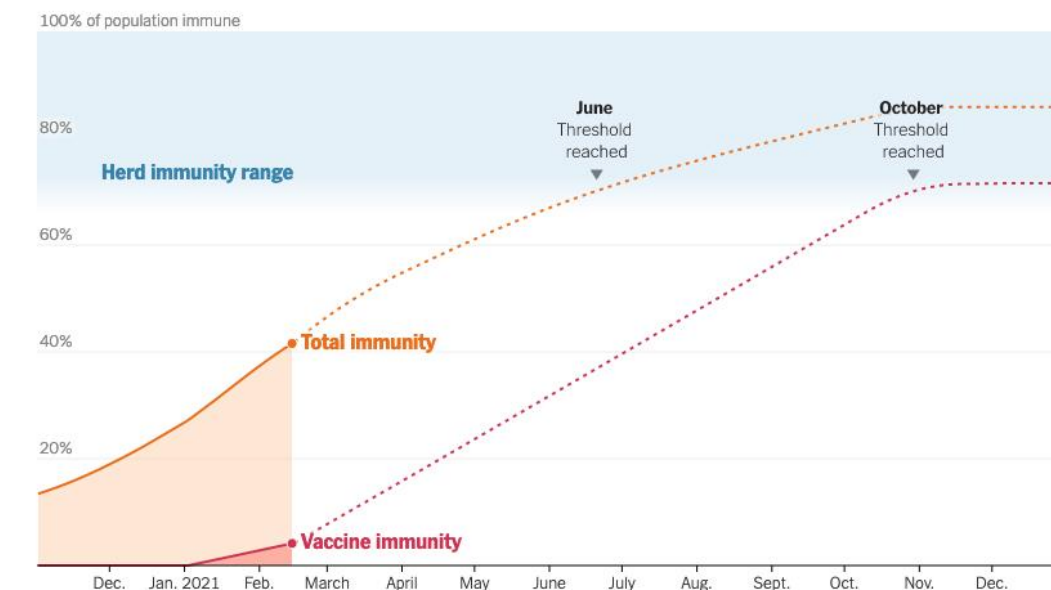
Design activity

While this means we would reach herd immunity sooner because more people would get sick and develop antibodies, that speed would come at a grim cost.

CHOOSE A SCENARIO 🗑

Keep measures in place Continue social distancing until most people are vaccinated.	Ease up in the spring Lift restrictions when 15 percent of people are vaccinated.	End precautions now Lift restrictions when few people are vaccinated.
---	---	---

If we end restrictions in April, we could reach the herd immunity threshold by **June**. But in that time, **170,000** people could die from the virus.



Precautions remain especially important as new variants of the virus emerge. If social distancing measures aren't followed, a stronger virus could rapidly infect and kill hundreds of thousands of people before they can be vaccinated.

What if a more contagious variant spreads?

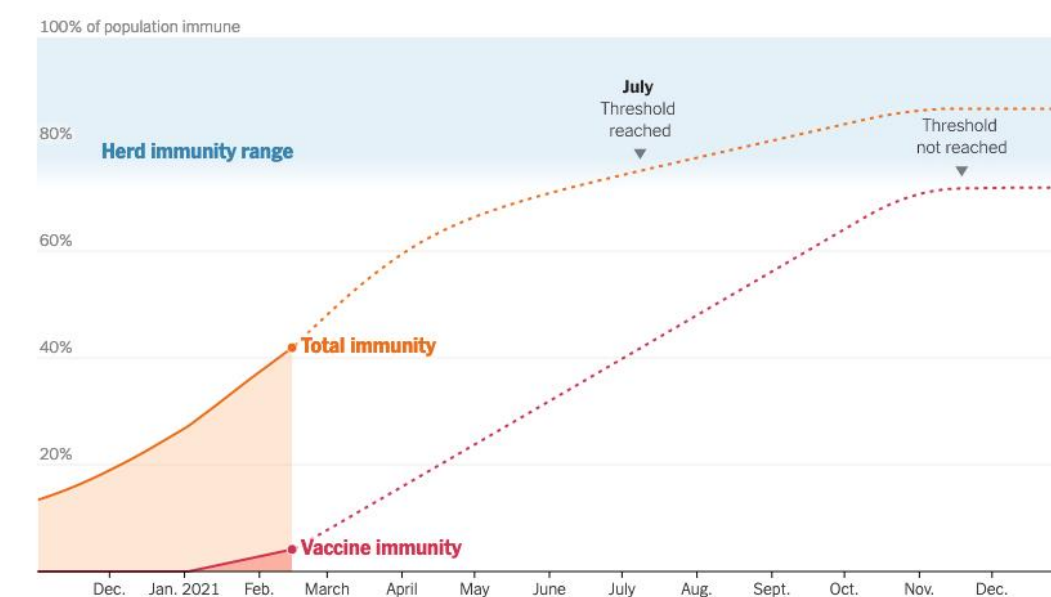
A more contagious variant of the virus, like the one first identified in Britain that is now [spreading](#) throughout the United States, could further complicate the path to herd immunity.

If the virus becomes more contagious, the threshold for herd immunity will go up. It may be hard for vaccines to keep pace, and precautions will be even more necessary to stem the spread.

CHOOSE A SCENARIO 🗑

Current variant with precautions Transmissibility and precautions remain the same.	More contagious variant with precautions Transmissibility increases but preventative measures continue.	More contagious variant with no precautions Preventative measures end in March and transmissibility increases.
--	---	--

A more contagious variant will spread faster and raise the herd immunity threshold. We could still expect to reach herd immunity in **July**, but new deaths could double, to **200,000** people.

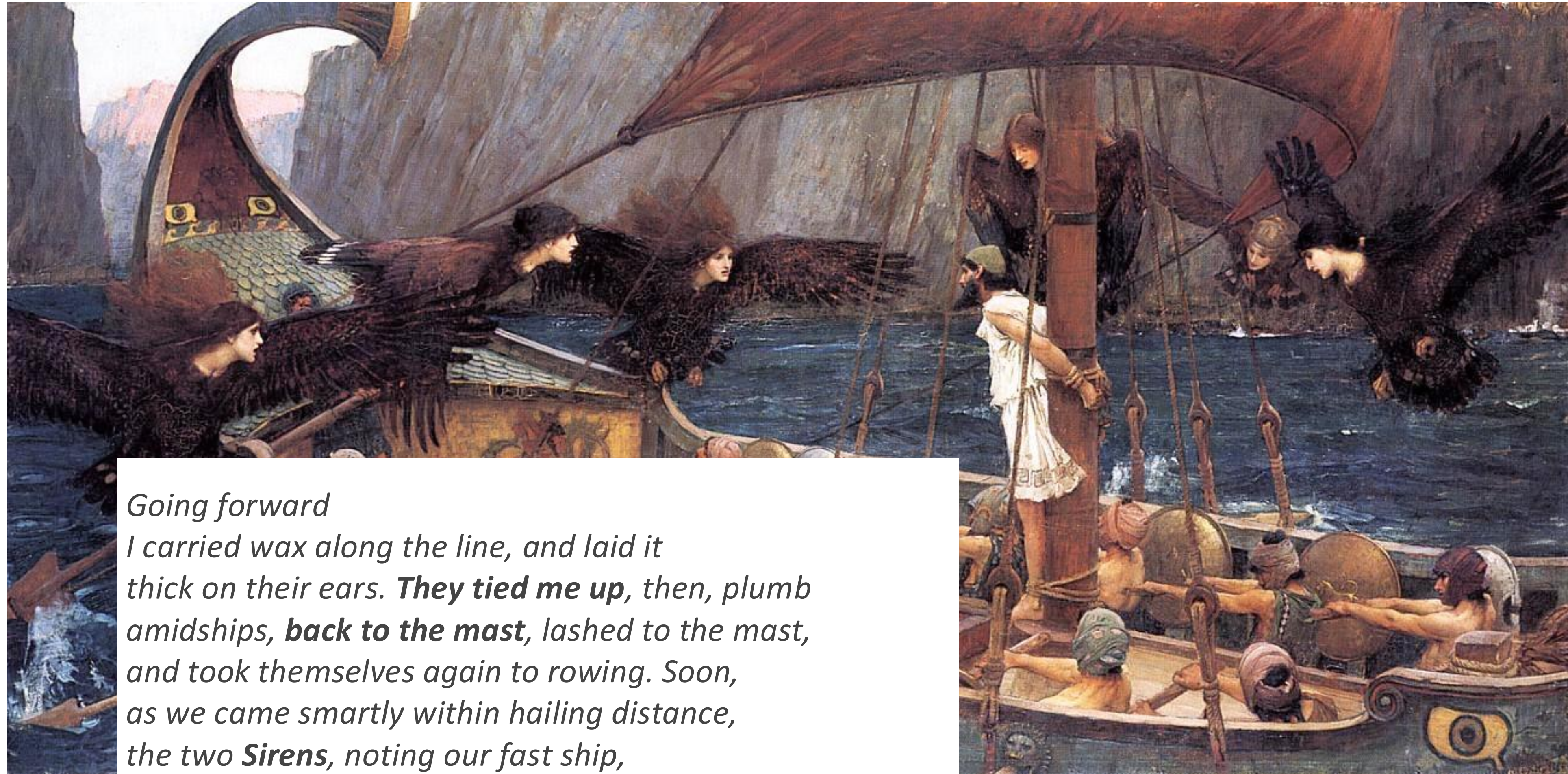


The spread of new virus variants makes it impossible to put a firm date on when we'll reach herd immunity or when the pandemic will end. There's a chance a mutation could lead to a version of the virus that doesn't respond to existing immunity, leading us to start the journey to herd immunity all over again.

Dr. Lee said that coronaviruses have relatively high mutation rates and that it is likely that new variants of the virus will continue to emerge. "The question will be how different might these variants be," he said. If a variant of the virus stops responding to the

STORYTELLING

as ancient as mankind

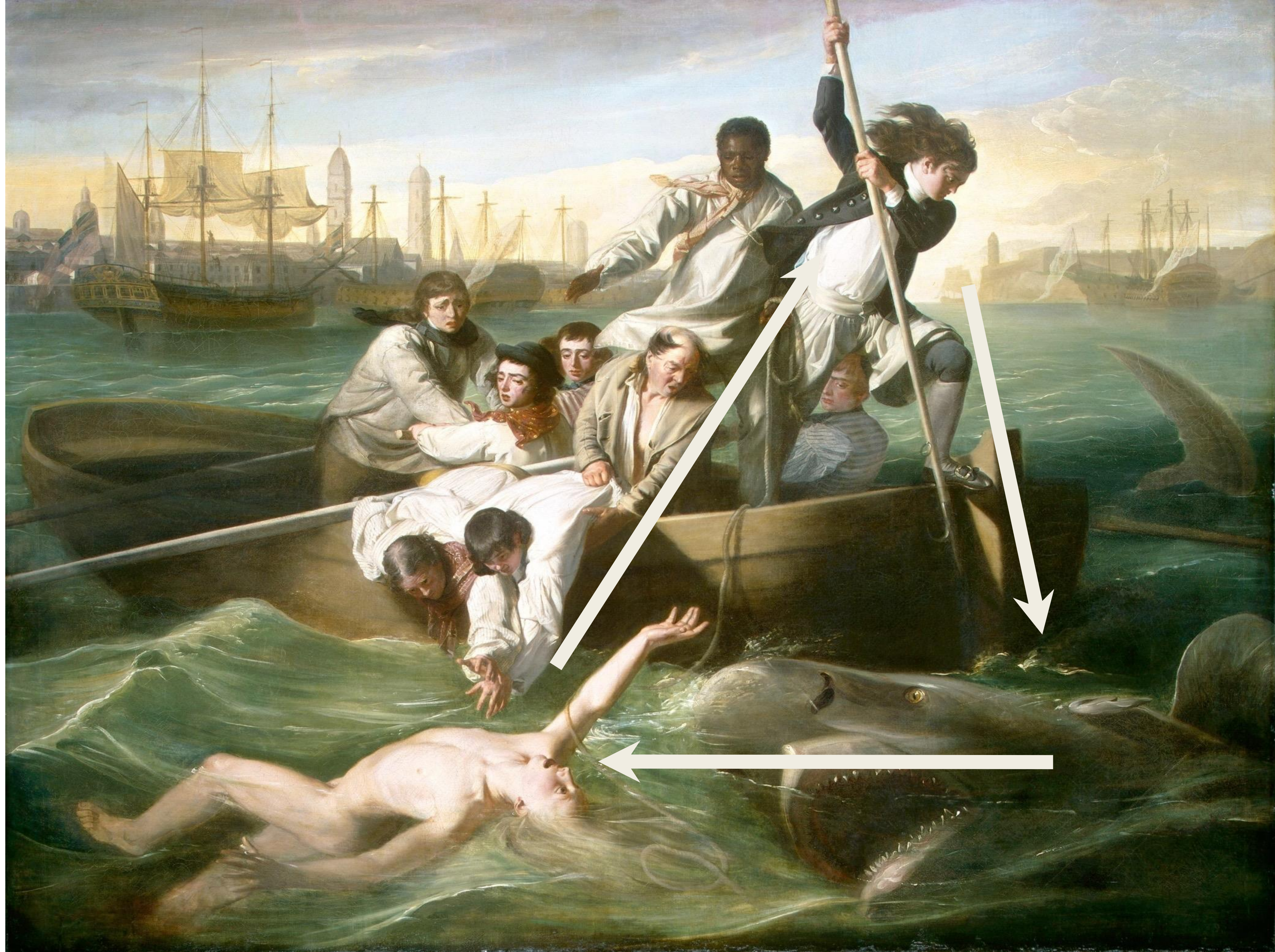


*Going forward
I carried wax along the line, and laid it
thick on their ears. **They tied me up**, then, plumb
amidships, **back to the mast**, lashed to the mast,
and took themselves again to rowing. Soon,
as we came smartly within hailing distance,
the two **Sirens**, noting our fast ship,
off their point, **made ready, and they sang...***

Narrative Storytelling

narrative (n): An account of a series of events, facts, etc., given in order and with the establishing of connections between them.

Effective storytelling “require[s] skills like those familiar to movie directors, beyond a technical expert’s knowledge of computer engineering and science.” - Gershon & Page ‘01







Justice Stephen G. Breyer to Retire From Supreme Court



Justice Stephen G. Breyer was appointed to the Supreme Court in 1994 by President

WASHINGTON — Justice Stephen G. Breyer, the senior member of the Supreme Court's three-member liberal wing, will retire, two people familiar with the decision said, providing President Biden a chance to make good on his campaign pledge to name a Black woman to the court.

Mr. Biden is expected to formally announce the retirement at the White House on Thursday, according to one person familiar with the planning for the event.

Justice Breyer, 83, the oldest member of the court, was appointed in 1994 by President Bill Clinton. After [the death of Justice Ruth Bader Ginsburg in 2020](#) and [the appointment of Justice Amy Coney Barrett](#) by President Donald J. Trump, he became the subject of an energetic campaign by liberals who wanted him to step down to ensure that Mr. Biden could name his successor while Democrats control the Senate.

Justice Stephen G. Breyer to Retire From Supreme Court



Justice Stephen G. Breyer was appointed to the Supreme Court in 1994 by President

WASHINGTON — Justice Stephen G. Breyer, the senior member of the Supreme Court’s three-member liberal wing, will retire, two people familiar with the decision said, providing President Biden a chance to make good on his campaign pledge to name a Black woman to the court.

Mr. Biden is expected to formally announce the retirement at the White House on Thursday, according to one person familiar with the planning for the event.

Justice Breyer, 83, the oldest member of the court, was appointed in 1994 by President Bill Clinton. After [the death of Justice Ruth Bader Ginsburg in 2020](#) and [the appointment of Justice Amy Coney Barrett](#) by President Donald J. Trump, he became the subject of an energetic campaign by liberals who wanted him to step down to ensure that Mr. Biden could name his successor while Democrats control the Senate.

Lede
(lead paragraph)

Justice Stephen G. Breyer to Retire From Supreme Court



Justice Stephen G. Breyer was appointed to the Supreme Court in 1994 by President

WASHINGTON — Justice Stephen G. Breyer, the senior member of the Supreme Court's three-member liberal wing, will retire, two people familiar with the decision said, providing President Biden a chance to make good on his campaign pledge to name a Black woman to the court.

Mr. Biden is expected to formally announce the retirement at the White House on Thursday, according to one person familiar with the planning for the event.

Justice Breyer, 83, the oldest member of the court, was appointed in 1994 by President Bill Clinton. After [the death of Justice Ruth Bader Ginsburg in 2020](#) and [the appointment of Justice Amy Coney Barrett](#) by President Donald J. Trump, he became the subject of an energetic campaign by liberals who wanted him to step down to ensure that Mr. Biden could name his successor while Democrats control the Senate.

**Nut Graf
(nutshell paragraph)**

STORYTELLING

across various media

PEOPLE TELL STORIES

WORDS TELL STORIES

IMAGES TELL STORIES

COMICS TELL STORIES

MOVIES TELL STORIES

DATA TELLS STORIES?





Privacy and the internet

Lives of others

BY ANDREW HARRIS

Facebook and Google face a backlash from users and regulators alike as for the way they have handled sensitive data

JENNIFER STUCKERT, Canada's privacy commissioner, infatuated with Facebook in August 2009, the social networking site struck a deal, agreeing to change its policies. Within a year it complied with the country's privacy law. Now even Ms Stuckert, the authority, appears to be seeing an important part of the deal, which included giving users a clear and easy-to-implement choice over whether to share private data with third parties. "It doesn't seem to me that Facebook is going in the right direction on this issue," she says, hinting that, without a change of course, the firm could soon become the subject of another formal investigation by her organisation.

Facebook is not the only internet giant to provide the best of data watching. Google ended advertising contracts this week following news that it had scanned some personal communications sent over its network. With data networks in homes and offices in some 30 countries. On May 17th Peter Schme, Germany's federal commissioner for data protection, called for an independent investigation into Google's behaviour, claiming that it had "simply disregarded normal rules in the development and usage of software."

The cases highlight rising tension between guardians of privacy and internet firms. And they reflect concerns among web users about how private data has made public. Several prominent internet

tyrants such as Cory Doctorow, a science-fiction author, and Ian Japane, a politician, have abandoned Facebook. Sites such as JustFacebook.com are urging others to do so, too. On May 2nd for instance Facebook "banned".

This is unlikely to dig the momentum of Facebook, which is poised to claim half a billion members and which draws over seven million visitors a week to its site. Yet charts for users have been ranked at the company's headquarters in Silicon Valley, where bosses are mulling how best to respond. Several senior folk are now hinting that Facebook will soon roll out stricter privacy controls to make it easier to keep more data hidden. MySpace, a rival, is already making its controls stricter in an effort to woo disaffected Facebookers to its service.

A month later Facebook's handling of privacy has been hovering for years. In December the social network changed the default settings on its privacy controls so that individuals' personal information would be shared with "everyone" rather than "selected friends". Facebook argued this reflected a shift in society towards greater openness, and noted that users could still adjust privacy settings. Back again, but increased privacy activism lobbied for its reversal.

The switch should not have come as a surprise. Early on, many social networks

Also in this section:

- 68 Spring conspiracies
- 69 The recovery of General Motors
- 69 Business crime in China
- 70 Indian retail
- 70 Japan's drug firms on the move
- 71 Brazil's talacoma
- 71 Schumpeter: Dismantled

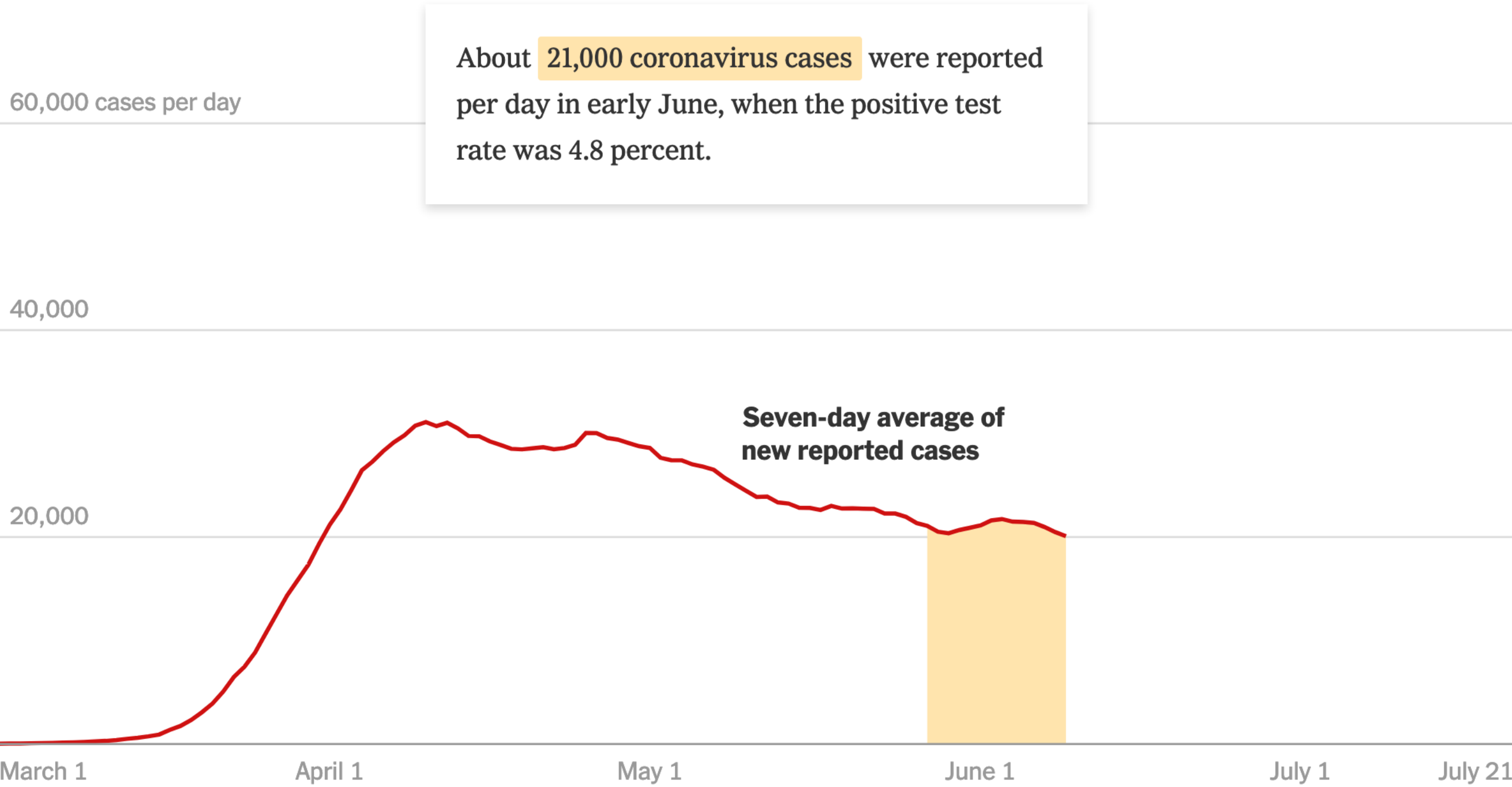
For daily analysis and news on business, visit economist.com/business-features

opened freely to such activity policies in order to attract and maintain users. But as major jobs controls are gradually loosened in emerging economies, the battle against which it sells advertising. And the more it knows about users' likes and dislikes, the better it can target ads that generate hundreds of millions of dollars.

Privacy grew freer still following a developers' conference last month at which Mark Zuckerberg, Facebook's boss, announced yet another series of policy changes. One that raised criticism was an "extended" password filter. Another that lets create third-party websites access Facebook data when people visit. Critics say that Facebook has made it sticky to delete this history, which may explain why its sudden policies have been so much.

Harquon of the site are gratifying to have Facebook too. This month a group of data-protection experts, also along the European Commission, wrote to the social network, calling its data on to become the default settings "unacceptable". And in the

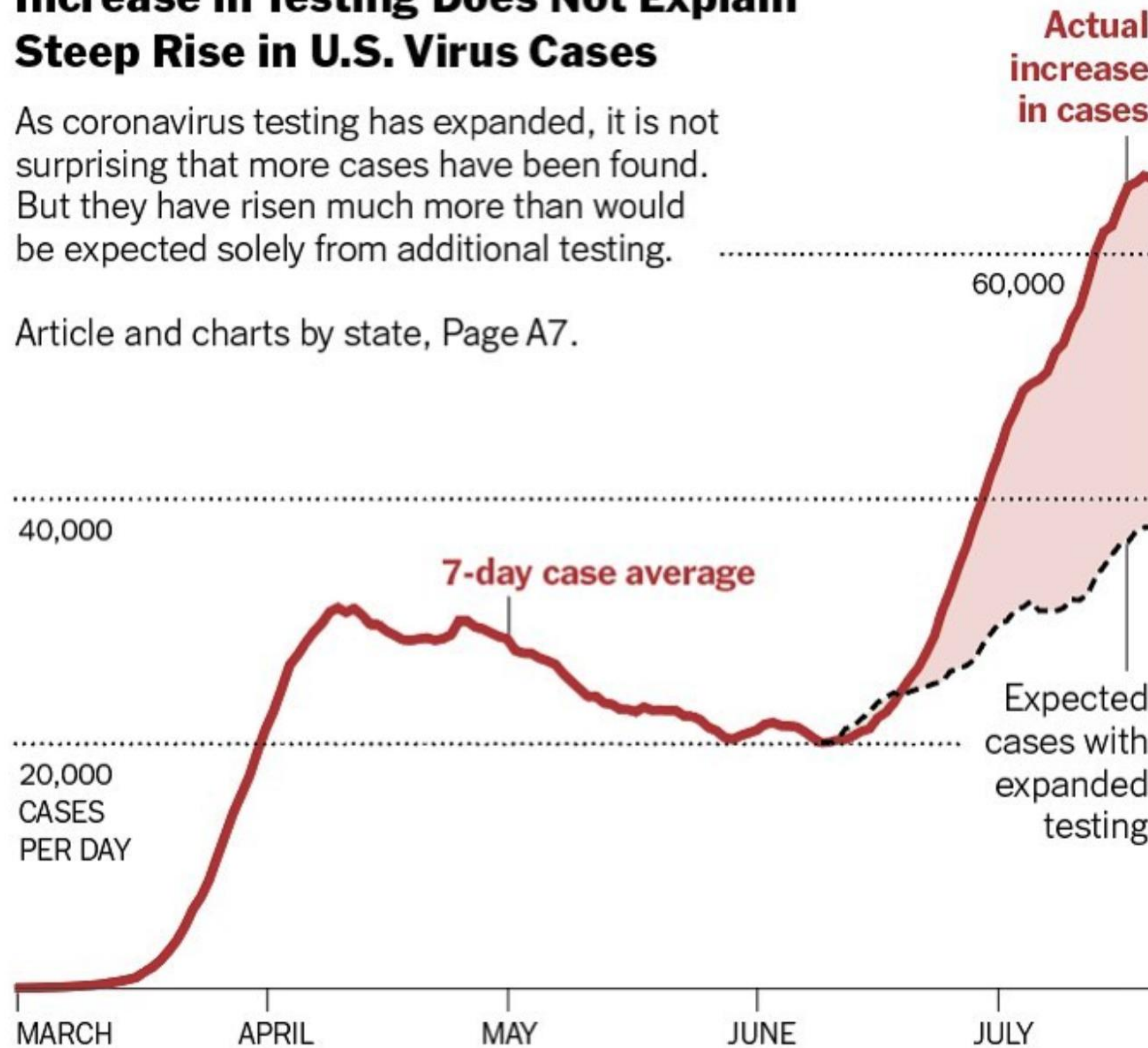




Increase in Testing Does Not Explain Steep Rise in U.S. Virus Cases

As coronavirus testing has expanded, it is not surprising that more cases have been found. But they have risen much more than would be expected solely from additional testing.

Article and charts by state, Page A7.



Source: The COVID Tracking Project

MATTHEW CONLEN/THE NEW YORK TIMES

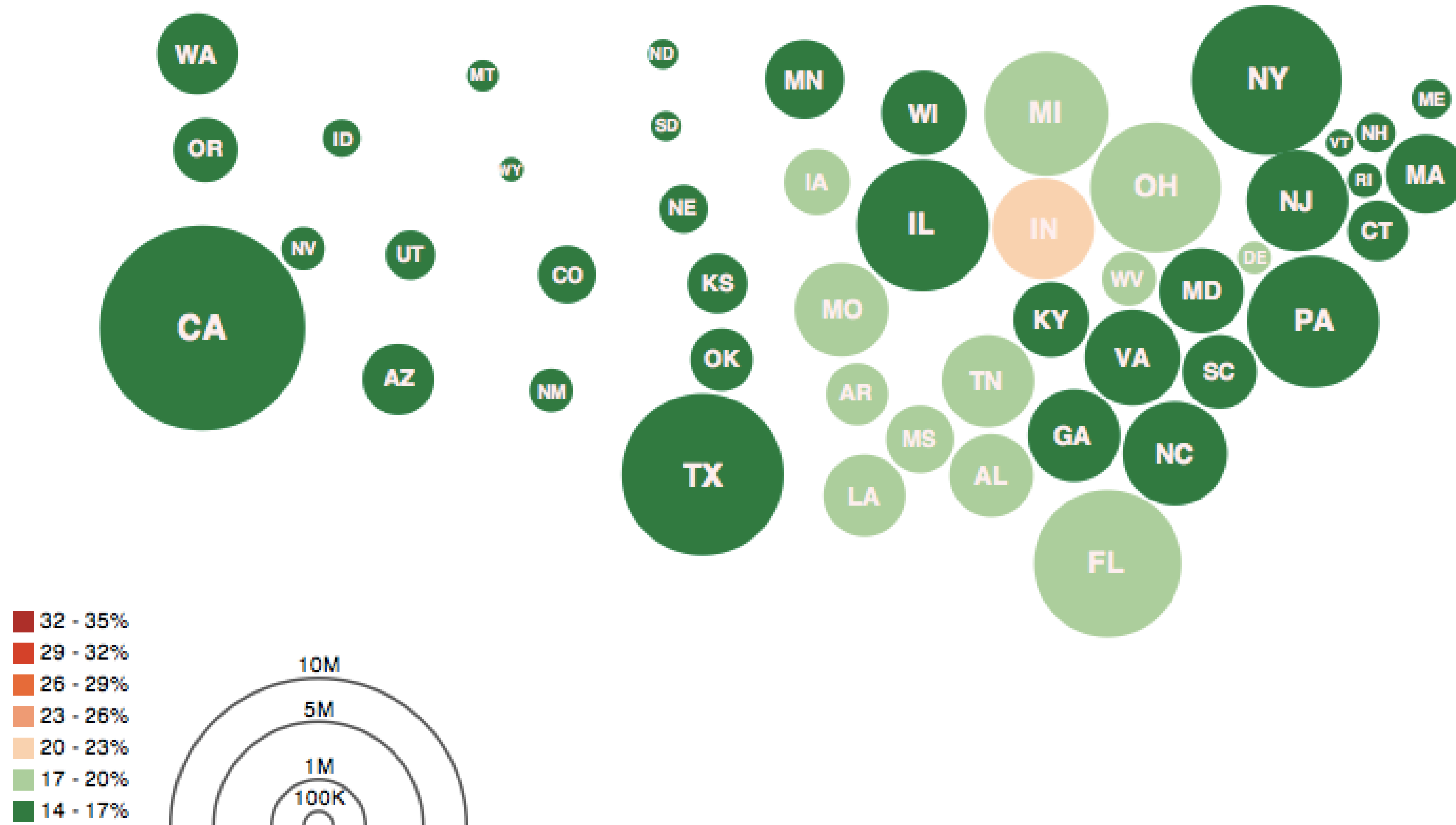
CASE STUDY COVID-19 TESTING

Input Data

Daily case and test counts.

Points of Focus

1. Baseline (smoothed average of daily cases)
2. Counterfactual case curve
(baseline positivity rate * daily test counts),
3. Actual case curve (daily cases)



Obesity Map Vadim Ogievetsky
1995



Of the 7,666 times that police officers killed people in the U.S. between 2013 and 2019...
25 (0.3%) resulted in a conviction



74 (1.0%) resulted in a charge but no conviction >



and...



monachalabi • Follow

United States



monachalabi Total disbelief about the system doing the bare minimum. George Floyd deserved so much more than this, he deserved to live.

Some detail about those 25 sentences:

Unknown sentence = 4 police killings

Just probation = 3

3 months in jail = 1

1 year in jail, 3 years suspended = 1

1 year in prison = 1

18 months in prison = 1

2.5 years in prison = 1

4 years in prison = 1

5 years in prison = 1

6 years in prison = 1

16 years in prison = 1

20 years in prison = 1

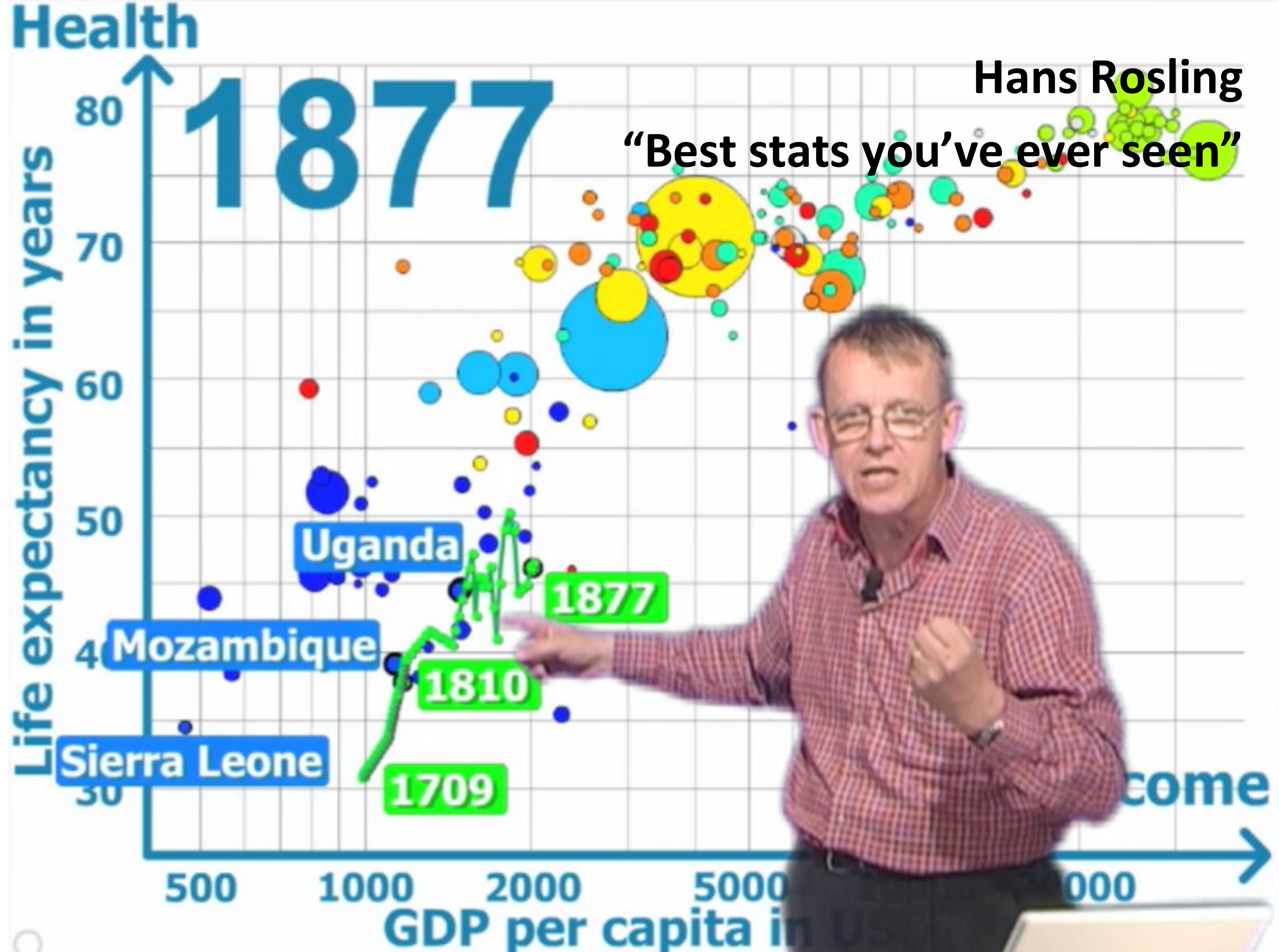
30 years in prison = 2



22,976 likes

3 HOURS AGO

Log in to like or comment.



Narrative Devices

What devices communicates best?

Highly dependent on: audience, context, format

Format

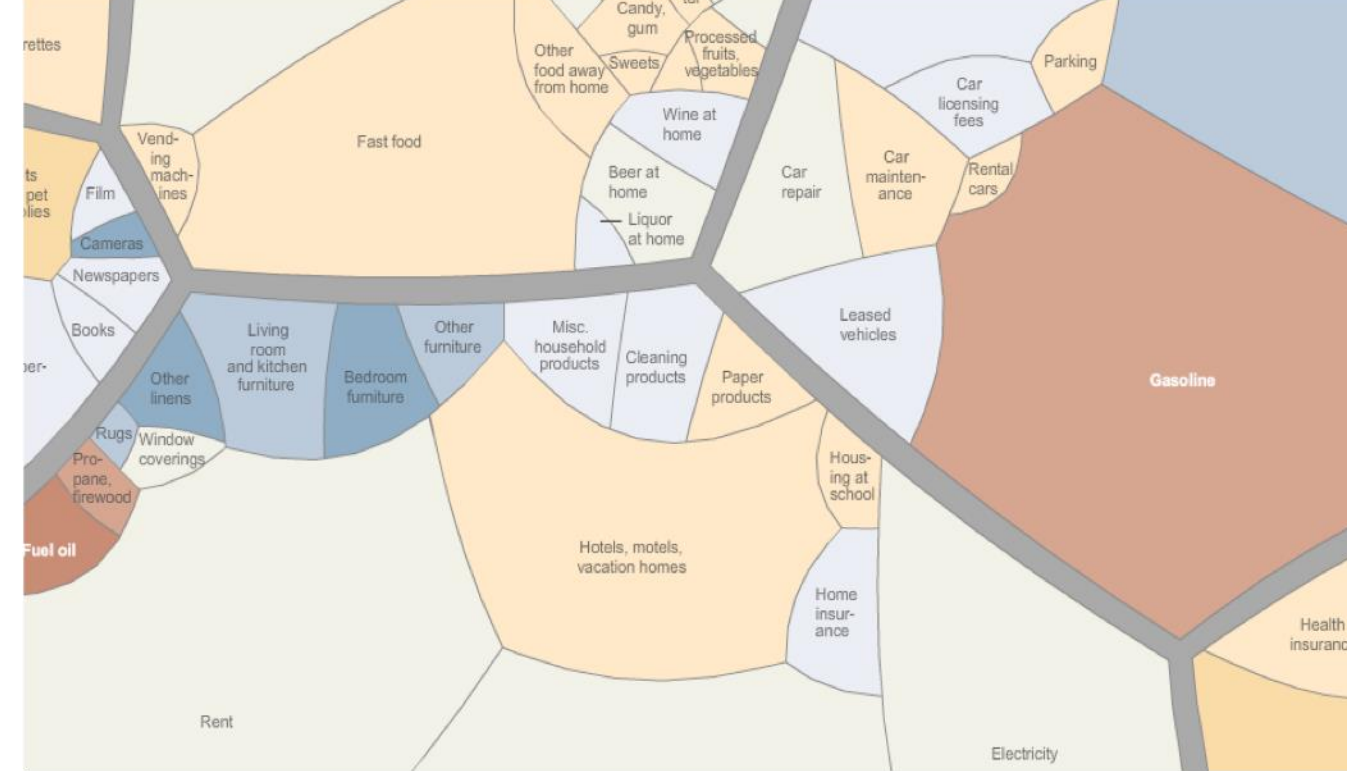
Interactive Article

Animation

Lecture

Still Image

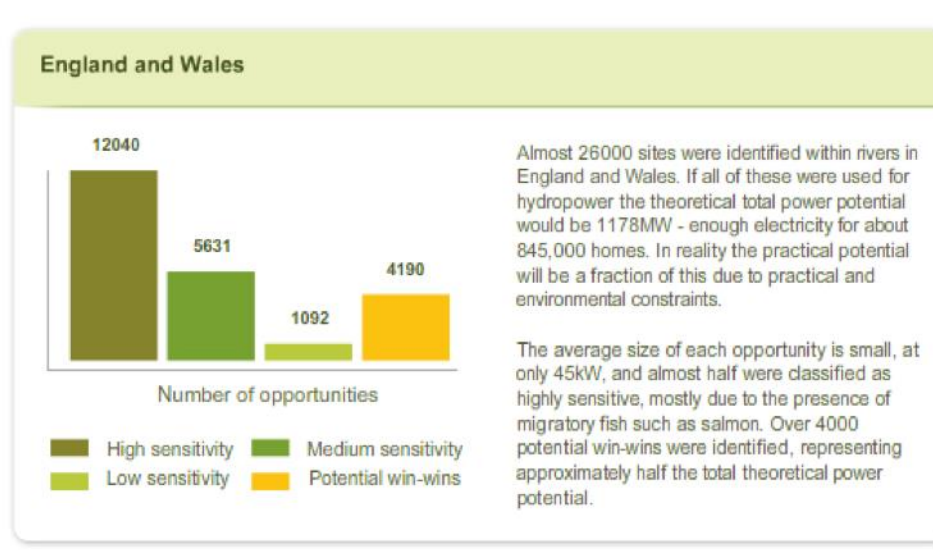
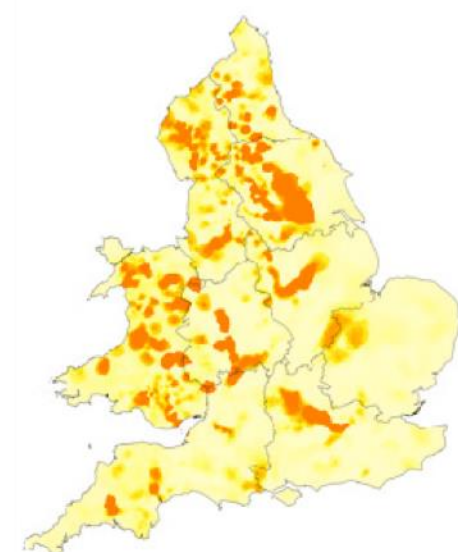
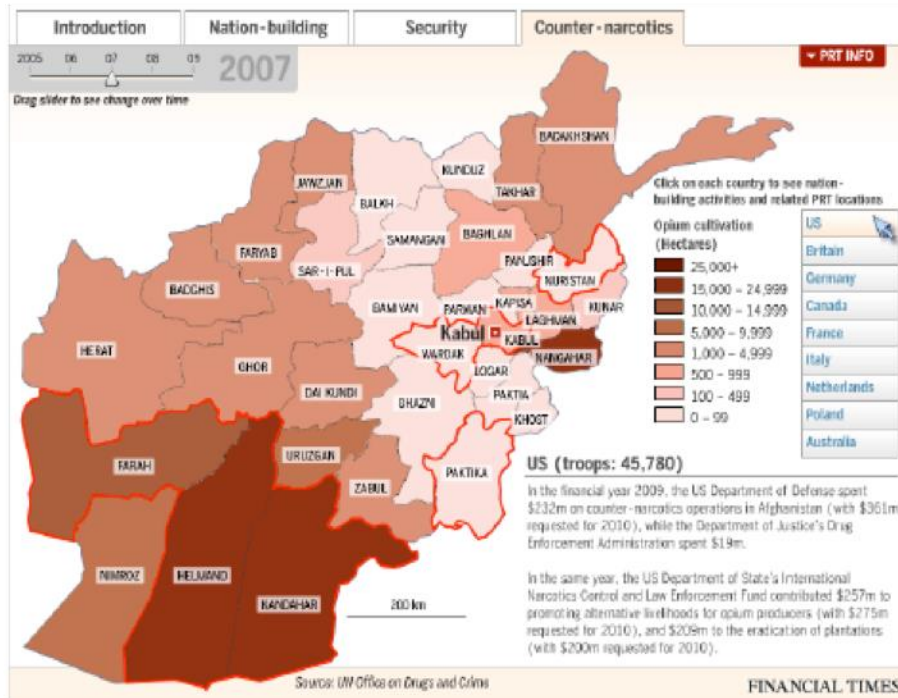
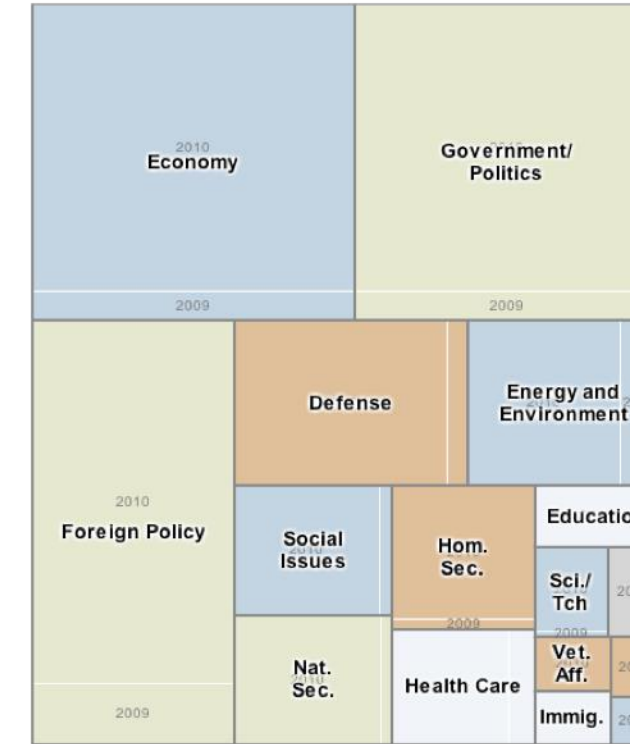
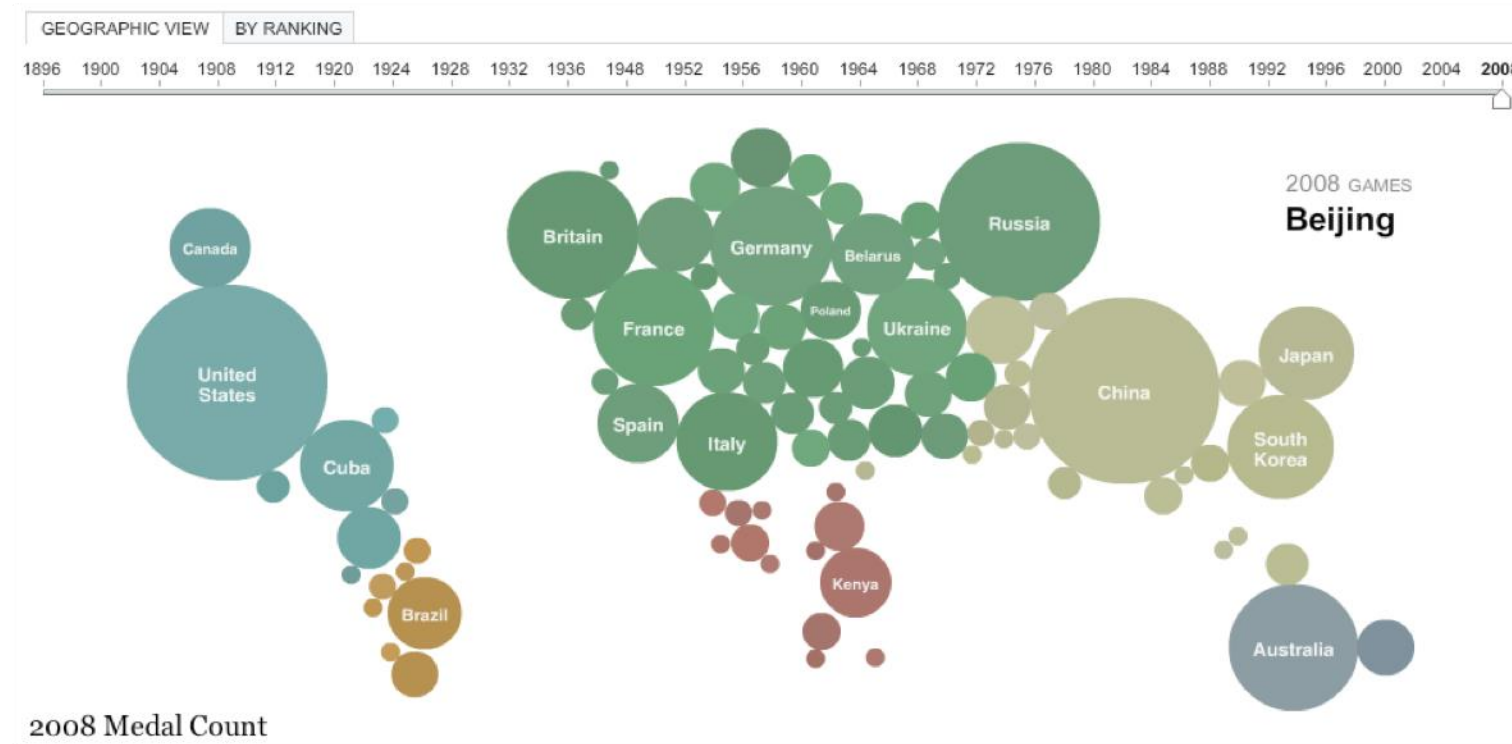
Video



58

CASE STUDIES

70% Journalism
20% Business
10% Research



Segel & Heer, 2010

755



Steroids or Not, the Pursuit Is On

Barry Bonds is taking aim at the career home run record. He needs only six more to tie Babe Ruth and 47 to equal Hank Aaron.

Lines are cumulative home runs.

Hank Aaron
755 homers
23 seasons



Babe Ruth
714 homers
22 seasons



Barry Bonds
708 homers
20 seasons

Bonds takes lead
Home runs
after 16 seasons
Bonds 567
Aaron 554
Ruth 516

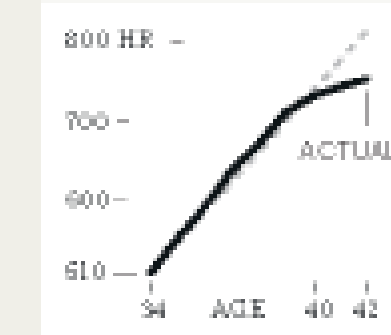
755
714
23 seasons
22 seasons
20 seasons
Bonds was injured last season. He played 14 games and hit 5 homers.

Homer Pace After Age 34

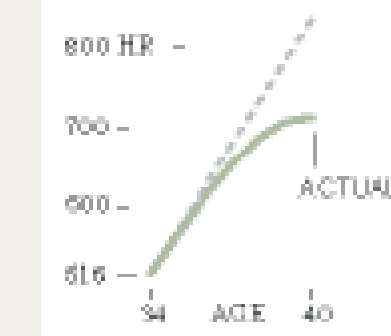
If the accusations are correct, Bonds was 34 in his first season on steroids. Here are projected home run paces for each player after age 34.

----- PROJECTED PACE BASED ON AVERAGE OF PREVIOUS FIVE SEASONS

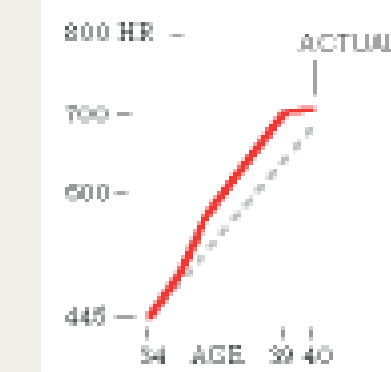
Aaron
Actual homers slightly outpace projected homers for five seasons.



Ruth
Averaged 46.4 homers a season from age 30 to 34. Averaged 42.5 for next four seasons.



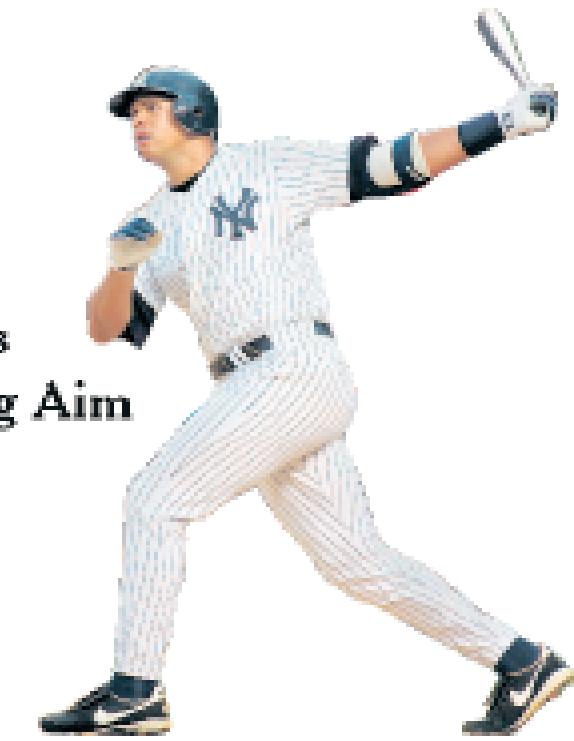
Bonds
From age 35 to 39, he averaged 14 more homers a season than projected.



Note: Ages as of July 1 of each season.

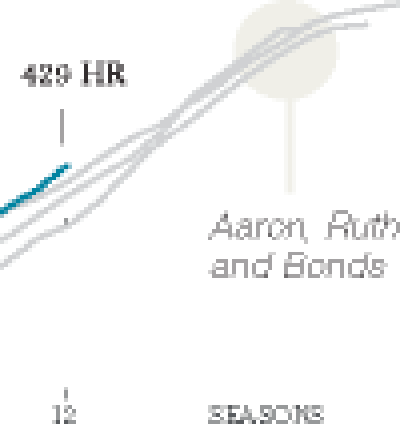
According to allegations in a book about Bonds, he began taking steroids before the 1999 season, his 14th in the league. Two seasons later, he hit 73 home runs, surpassing Aaron's career pace.

Others Taking Aim



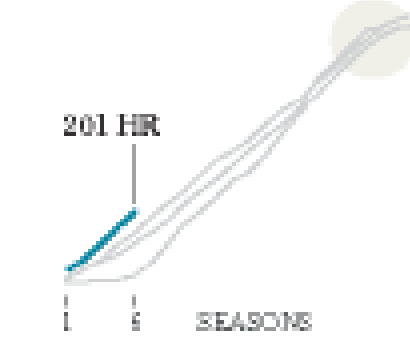
Alex Rodriguez

Is ahead of the pace set by all three home run leaders.



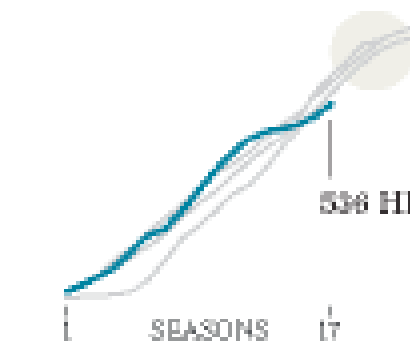
Albert Pujols

Averaging 40 homers a season, he has started stronger than the three leaders did.



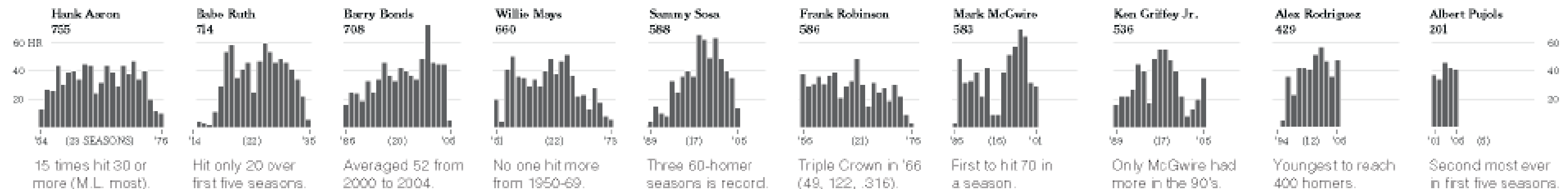
Ken Griffey Jr.

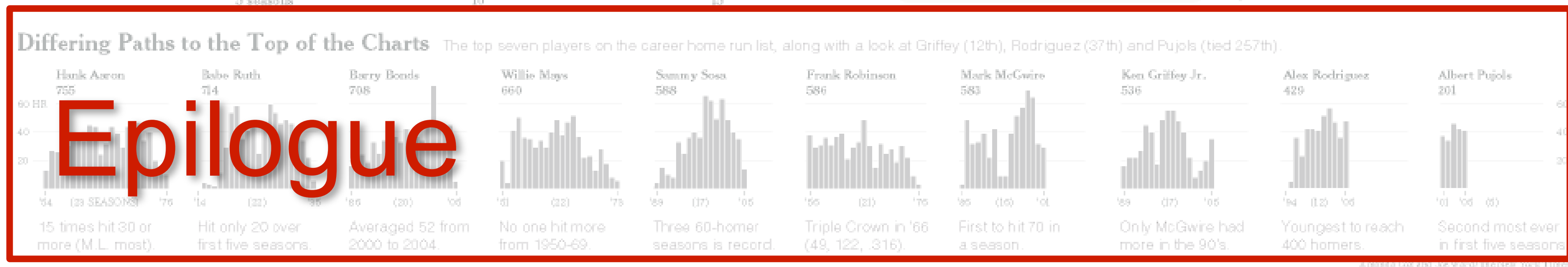
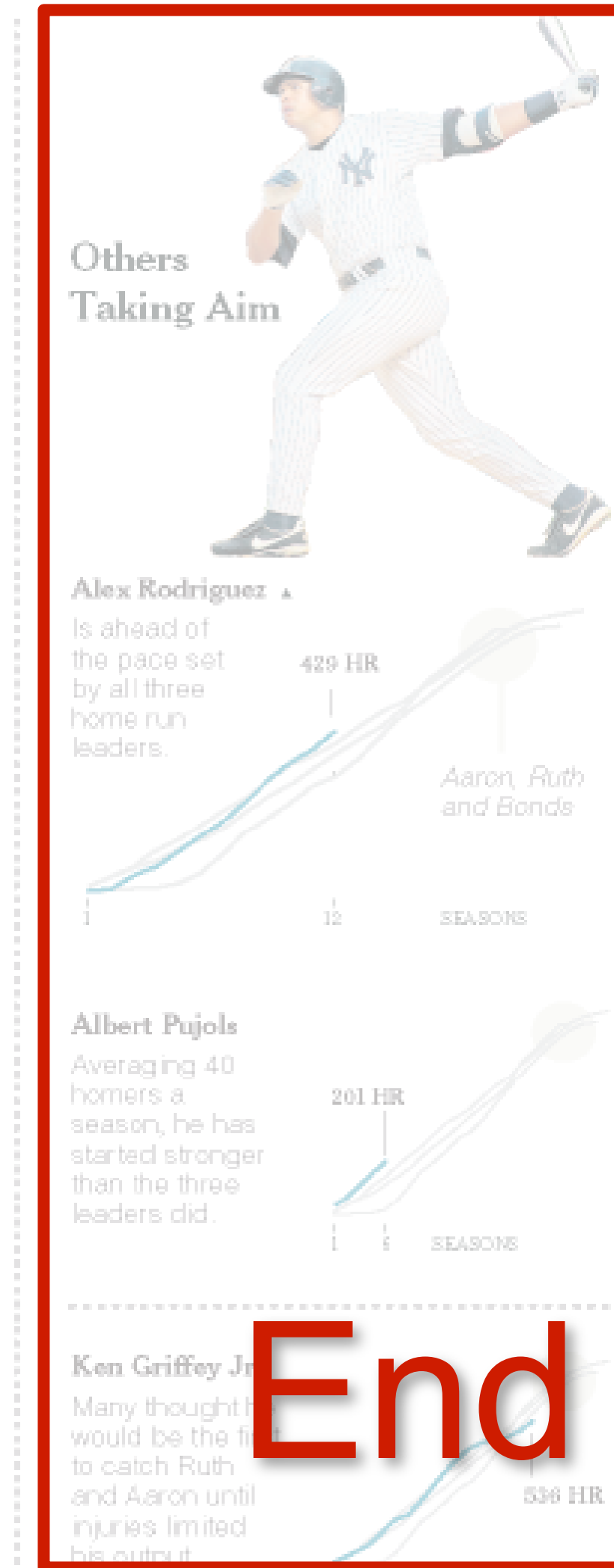
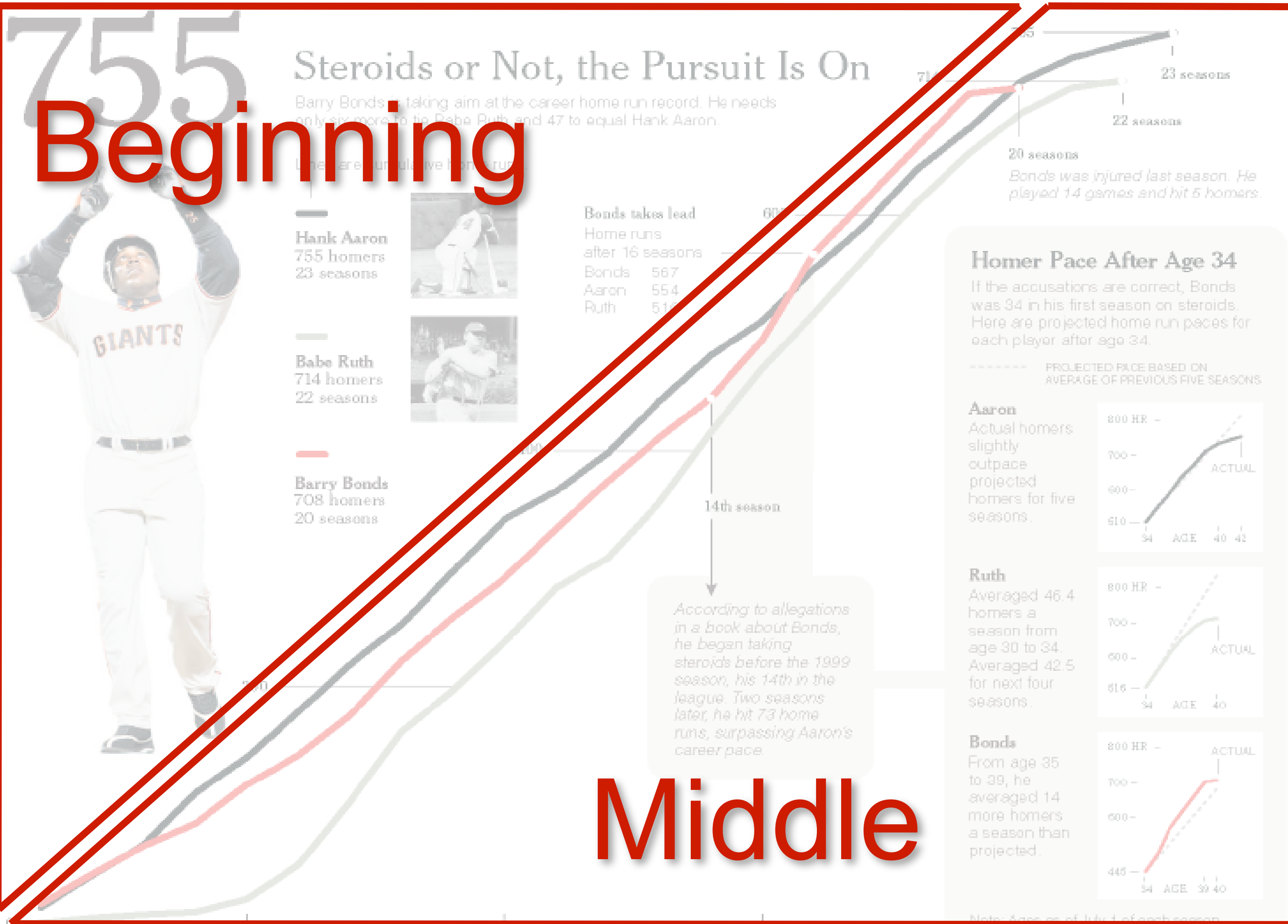
Many thought he would be the first to catch Ruth and Aaron until injuries limited his output.

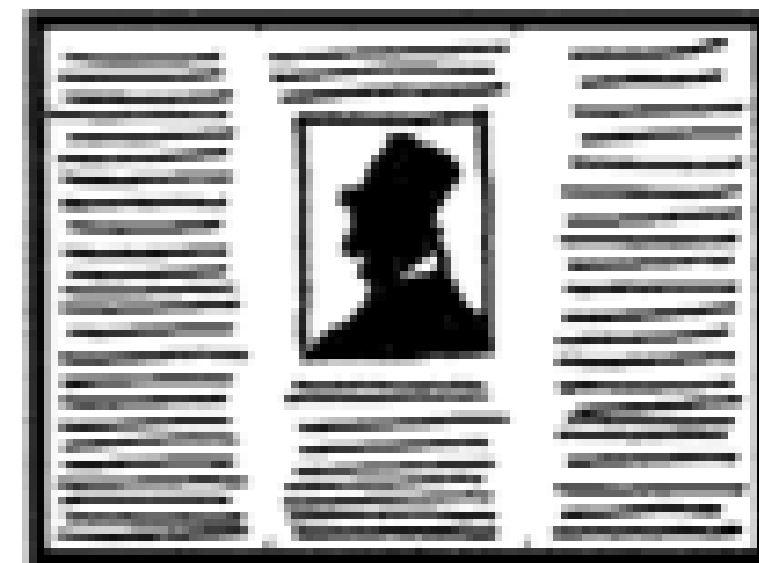


Differing Paths to the Top of the Charts

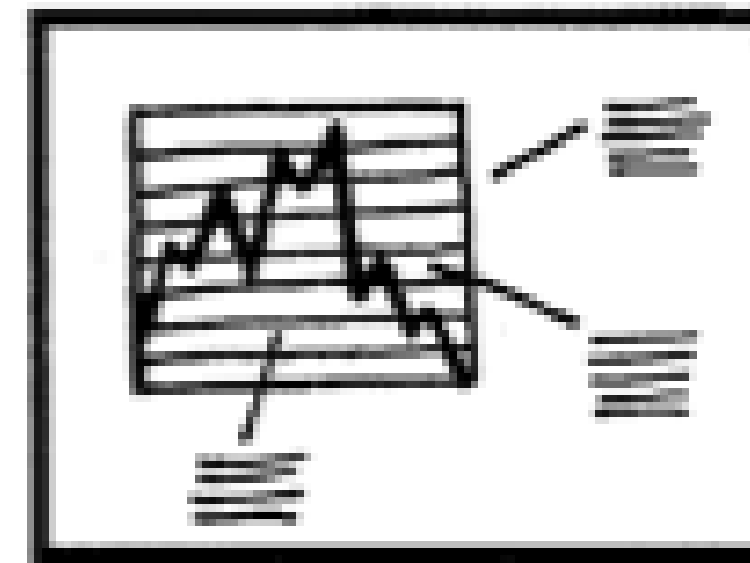
The top seven players on the career home run list, along with a look at Griffey (12th), Rodriguez (37th) and Pujols (tied 257th).



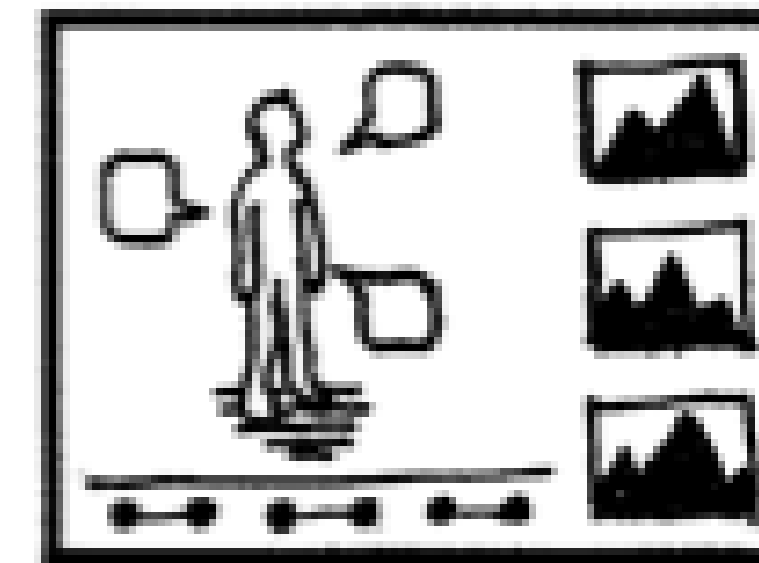




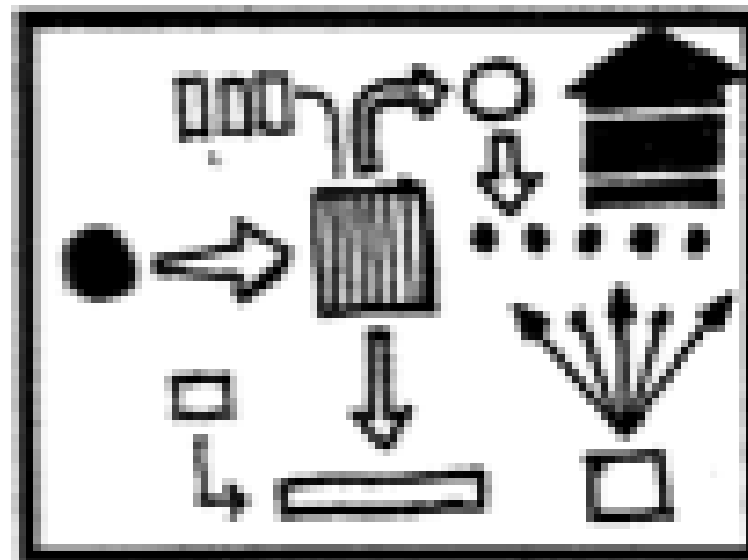
Magazine Style



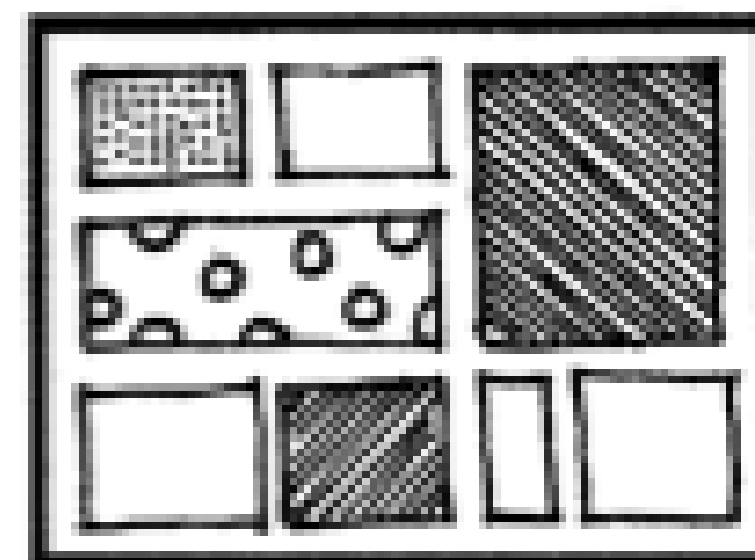
Annotated Chart



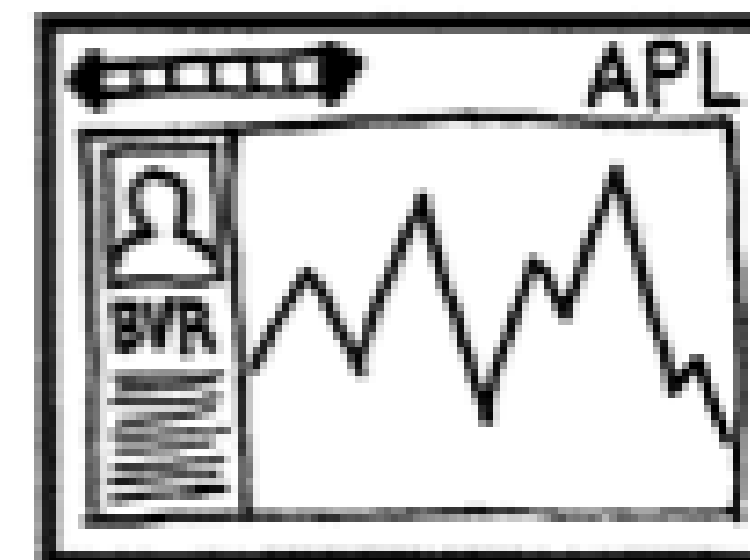
Partitioned Poster



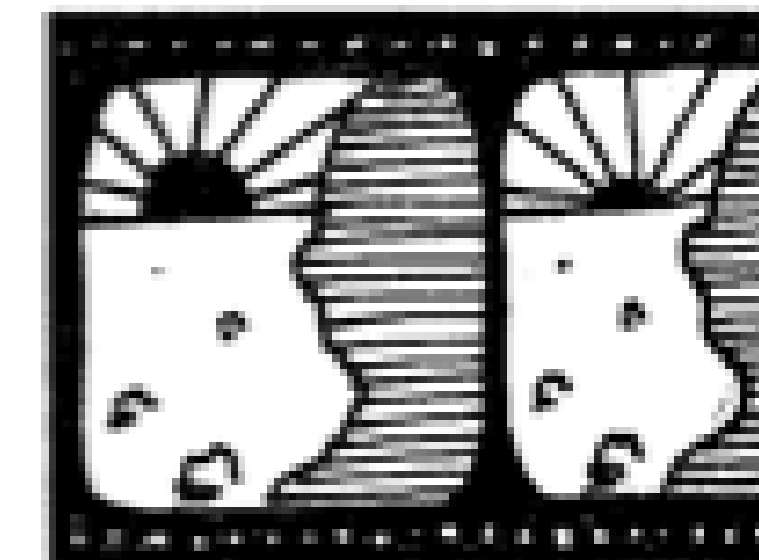
Flow Chart



Comic Strip



Slide Show

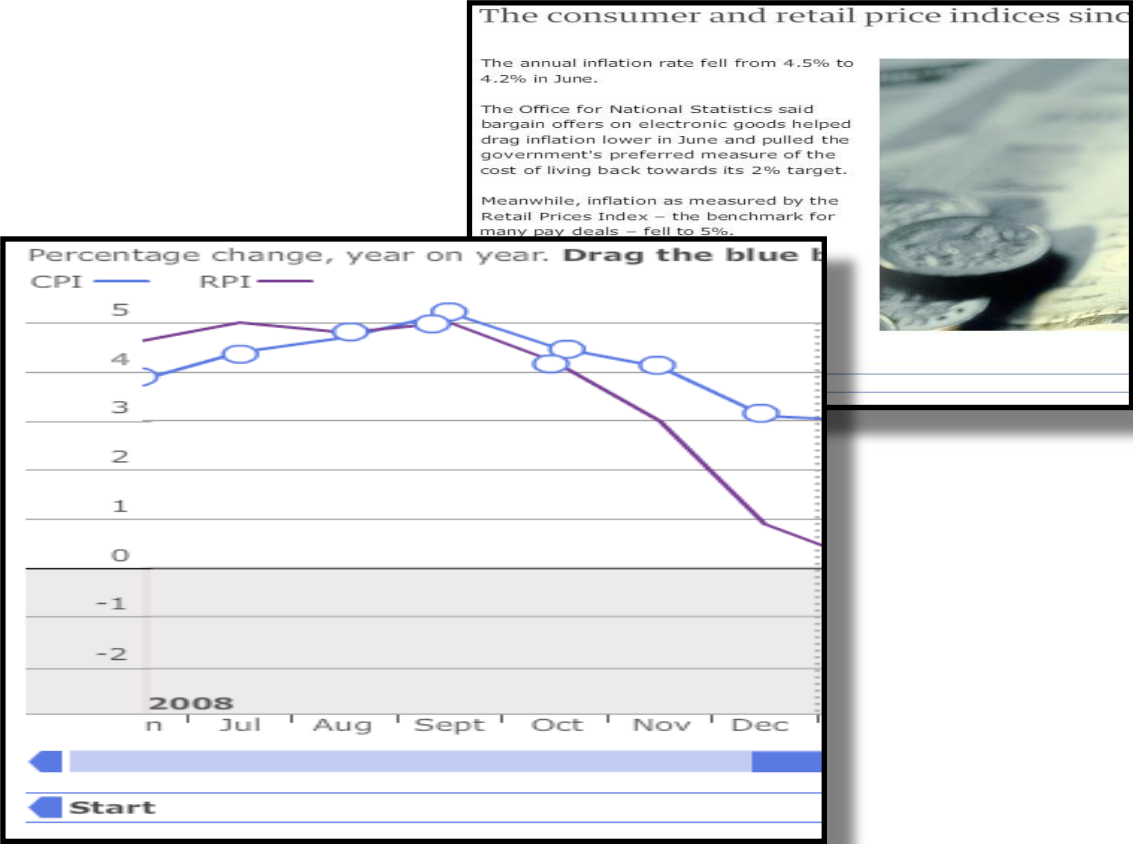


Film/Video/Animation

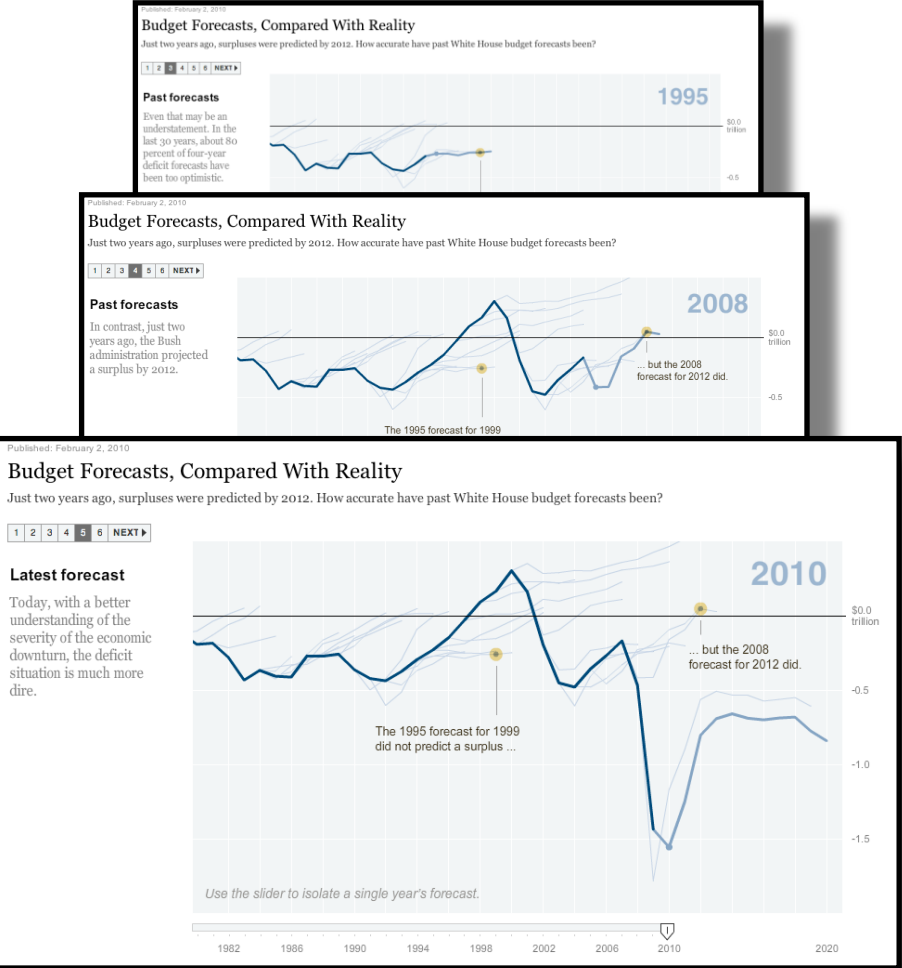
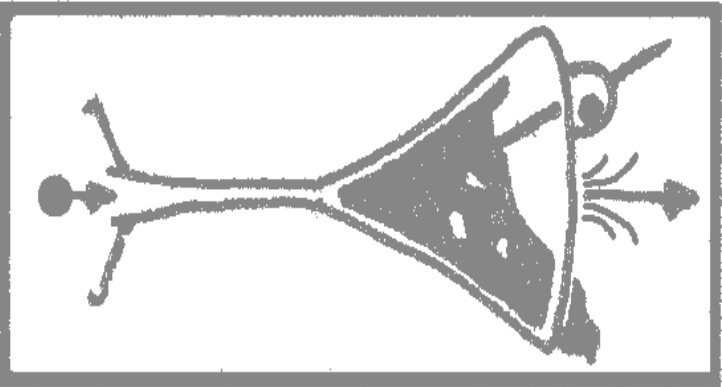
Genres for Narrative Visualization (2010)

Genres + Interactivity + Messaging =
DESIGN SPACE

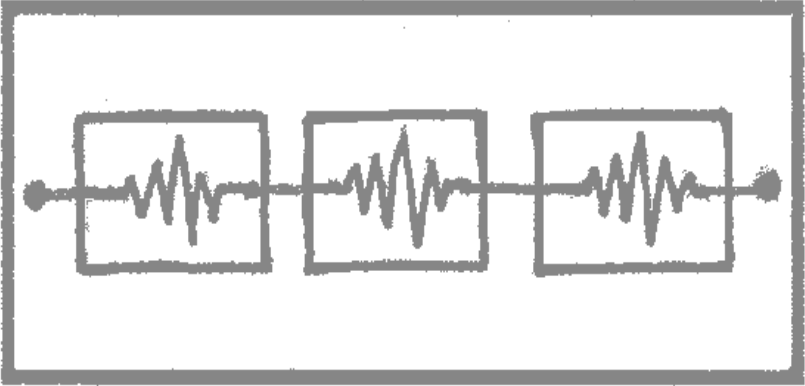
STORYTELLING	Author Driven strong ordering heavy messaging limited interactivity	Reader Driven weak ordering light messaging free interactivity	ASK QUESTIONS
CLARITY			EXPLORE
SPEED			FIND



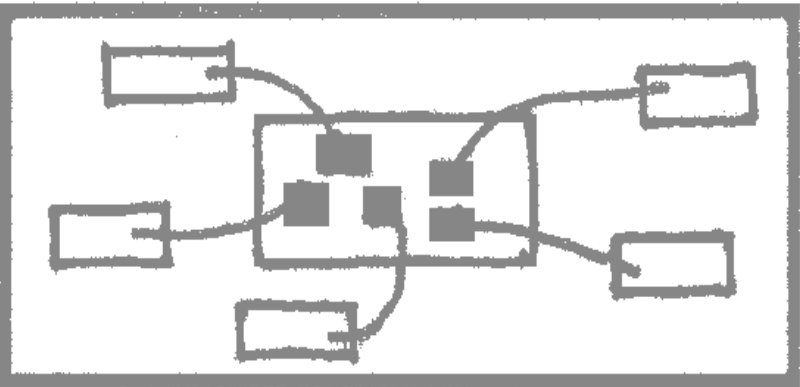
martini glass

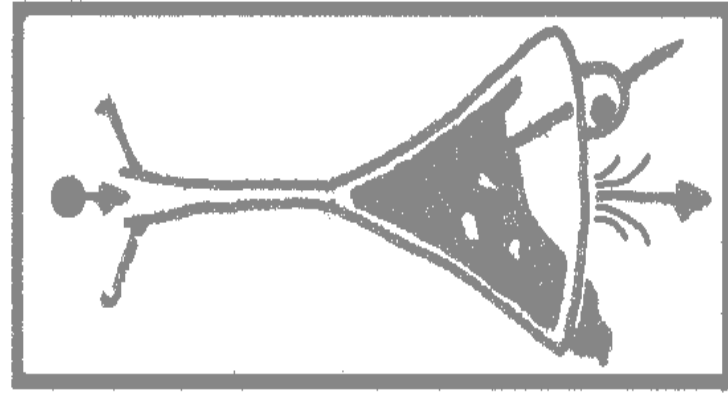


interactive
slideshow

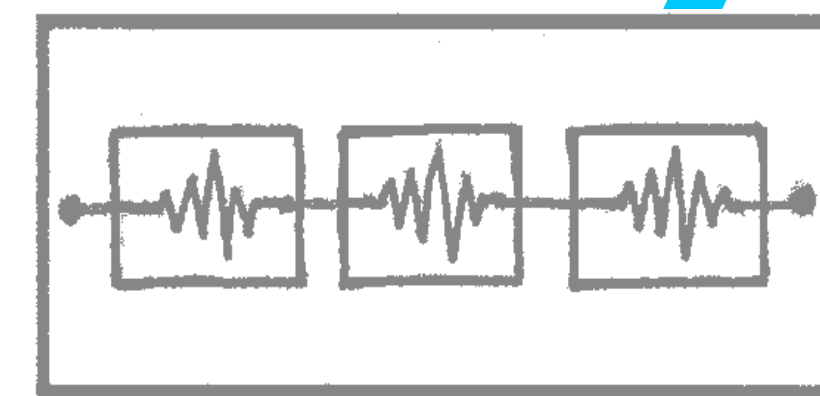
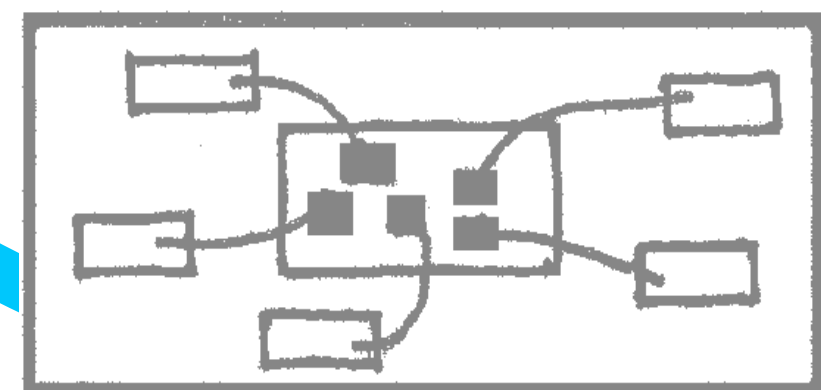
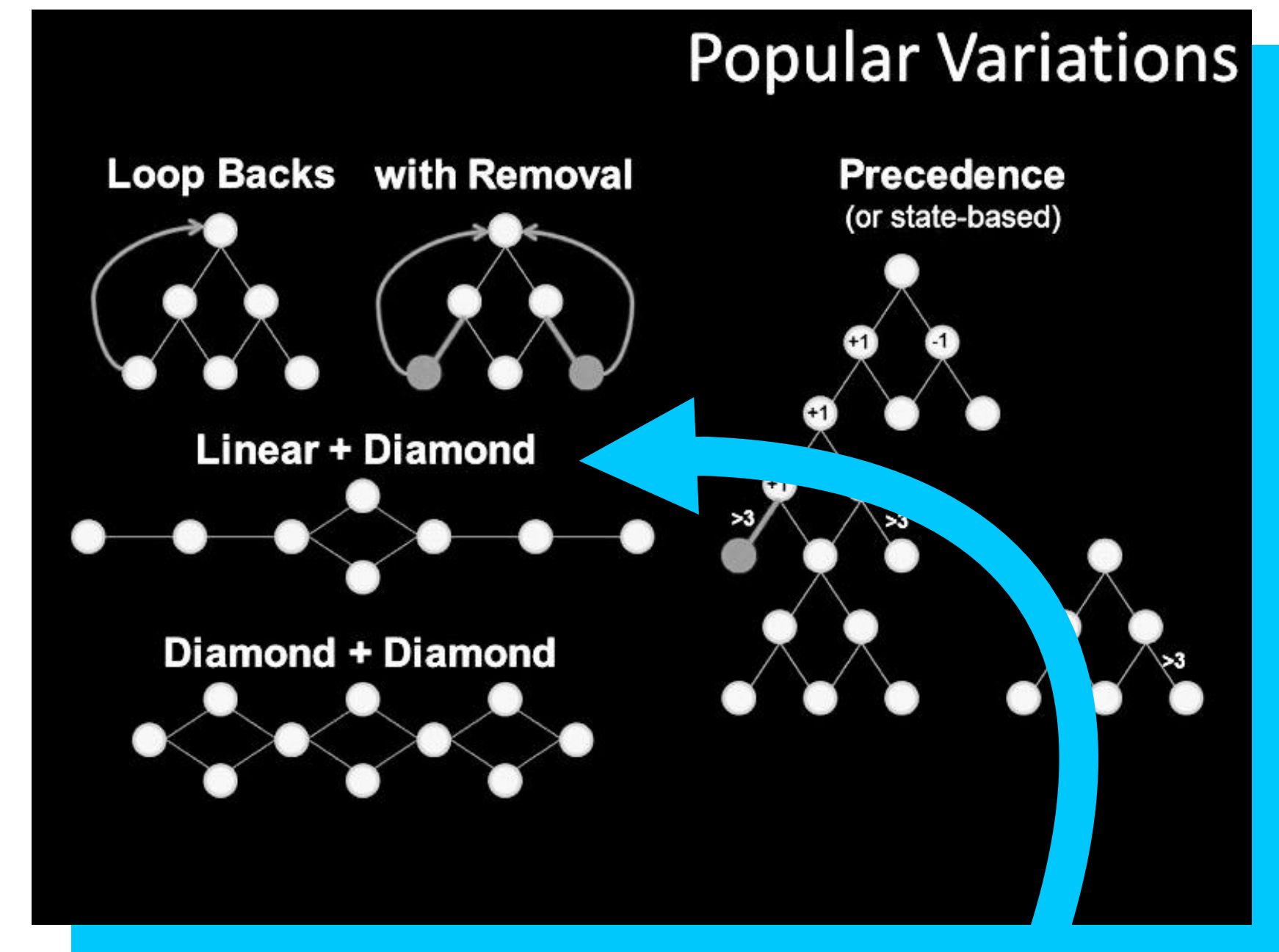
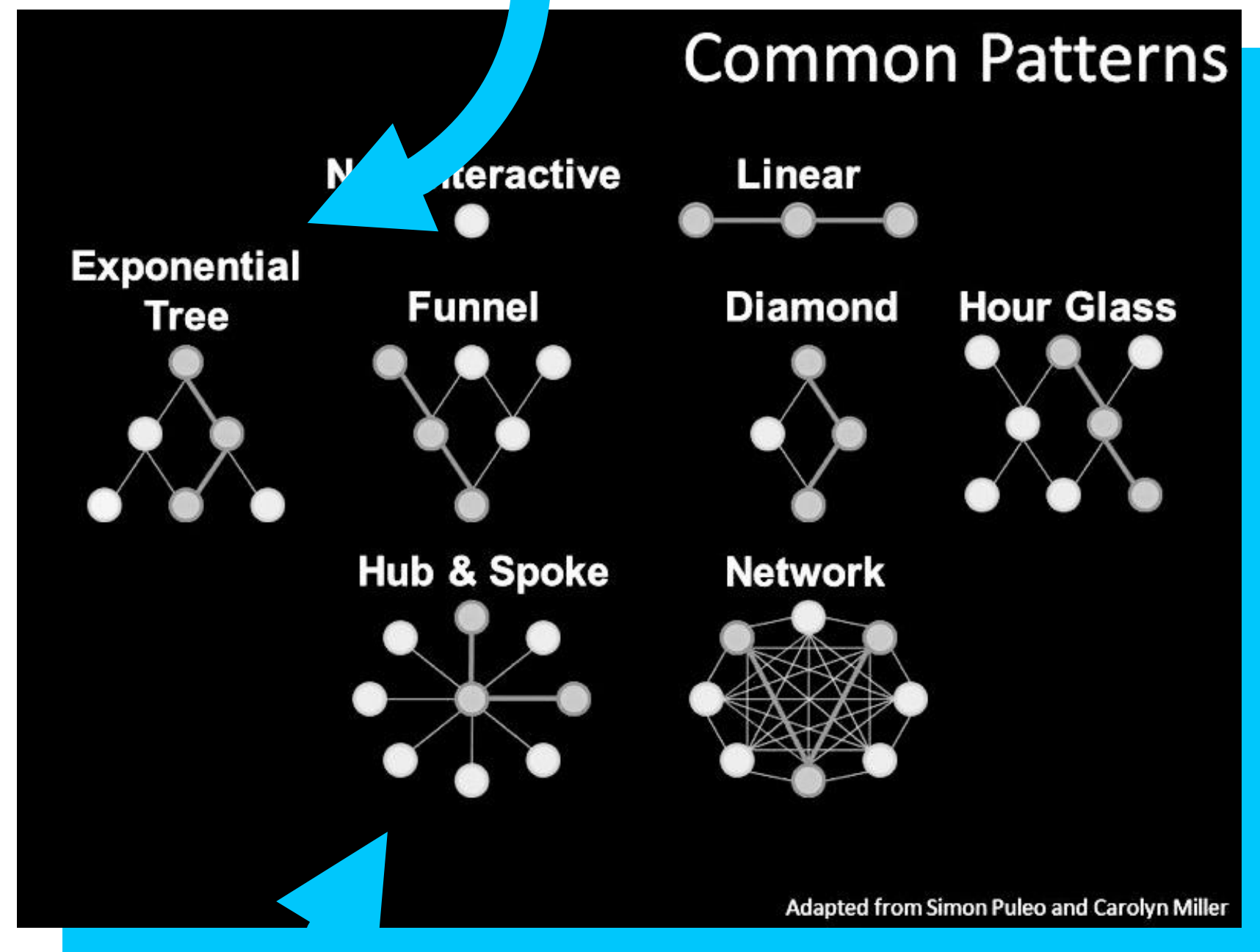


drill-down
story





(more patterns)

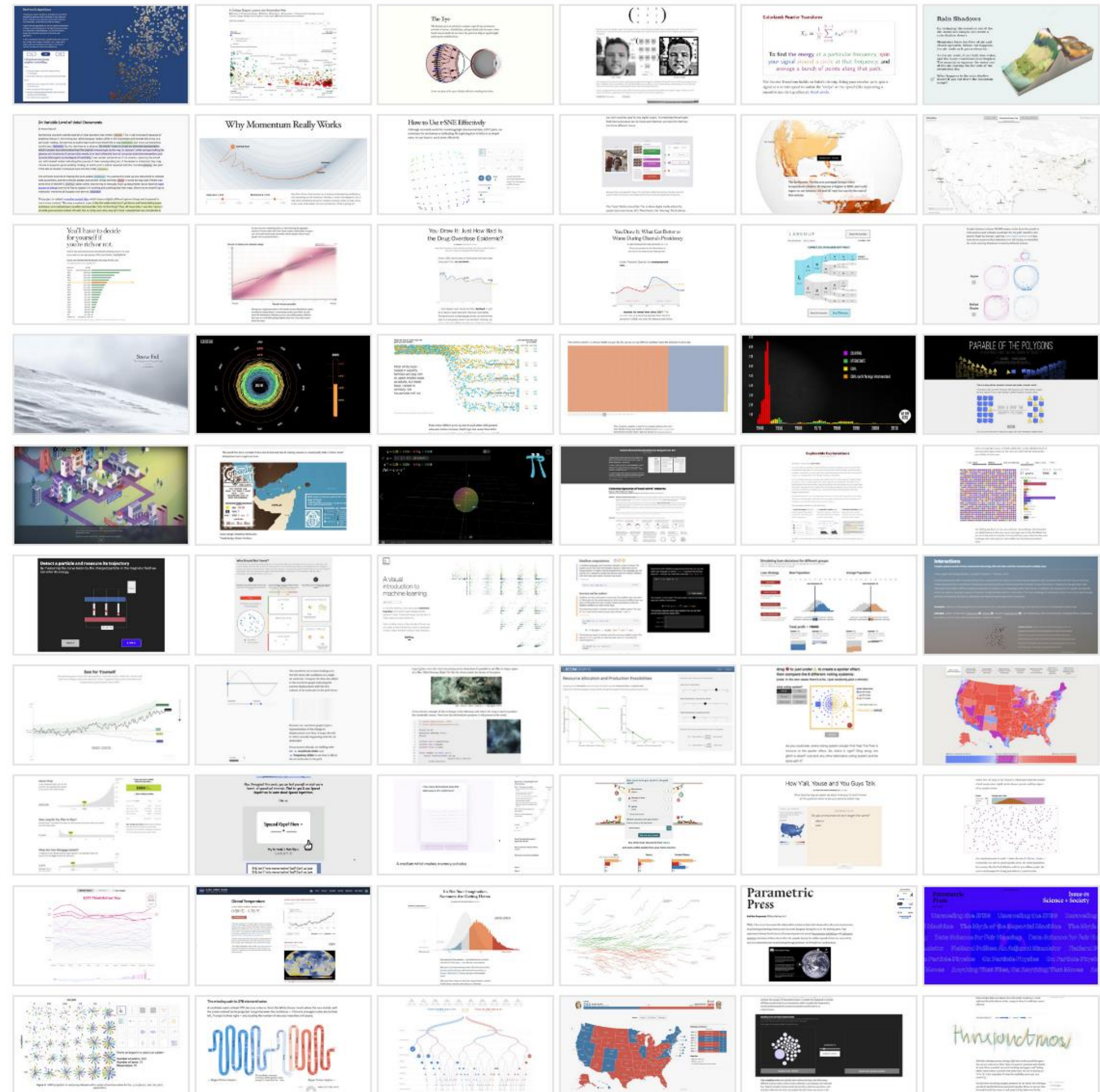


Communicating with Interactive Articles

60 Interactive Articles

NYTimes, WaPo, Distill, VisXAI, ...

Tied together research from HCI,
multimedia learning, infovis,
digital journalism



Hohman, Conlen, Heer, Chau, Distill, 2020

Interactive Articles - Design

Five affordances of the format

Connecting People and Data.

Make data pleasant to work with. Happy readers are engaged readers.

Personalizing Reading.

Let readers choose the content that is relevant to their own experience.

Making Systems Playful.

Run interactive simulations directly in the browser. No setup required.

Prompting Self-Reflection.

Help readers learn by asking them to reflect in a low pressure environment.

Reducing Cognitive Load.

Use effective representations to make complex topics more intuitive.

Interactive Articles - Design

Five affordances of the format

Connecting People and Data.

Make data pleasant to work with. Happy readers are engaged readers.

Making Systems Playful.

Run interactive simulations directly in the browser. No setup required.

Reducing Cognitive Load.

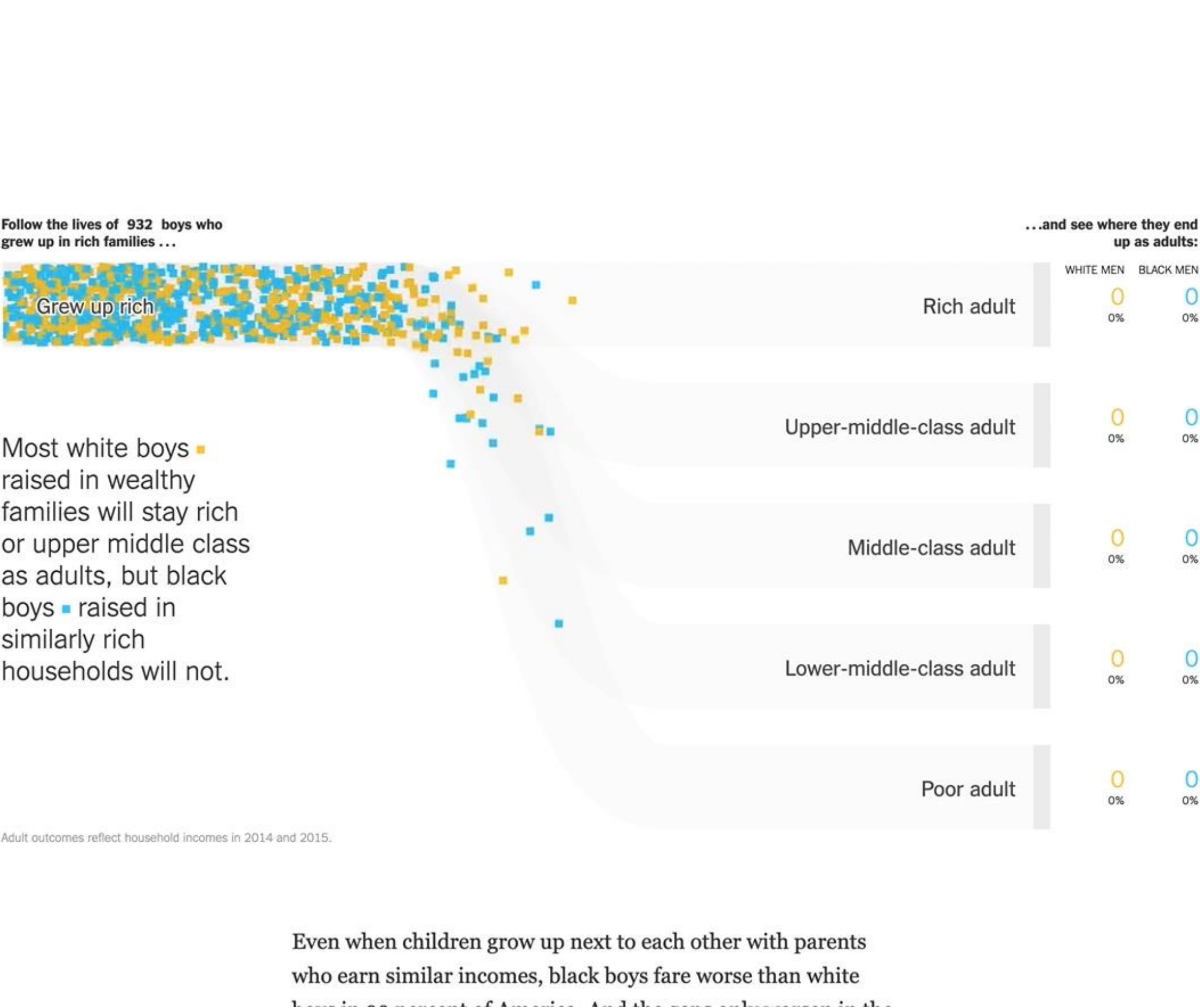
Use effective representations to make complex topics more intuitive.

Personalizing Reading.

Let readers choose the content that is relevant to their own experience.

Prompting Self-Reflection.

Help readers learn by asking them to reflect in a low pressure environment.



Interactive Articles - Design

Five affordances of the format

Connecting People and Data.

Make data pleasant to work with. Happy readers are engaged readers.

Personalizing Reading.

Let readers choose the content that is relevant to their own experience.

Making Systems Playful.

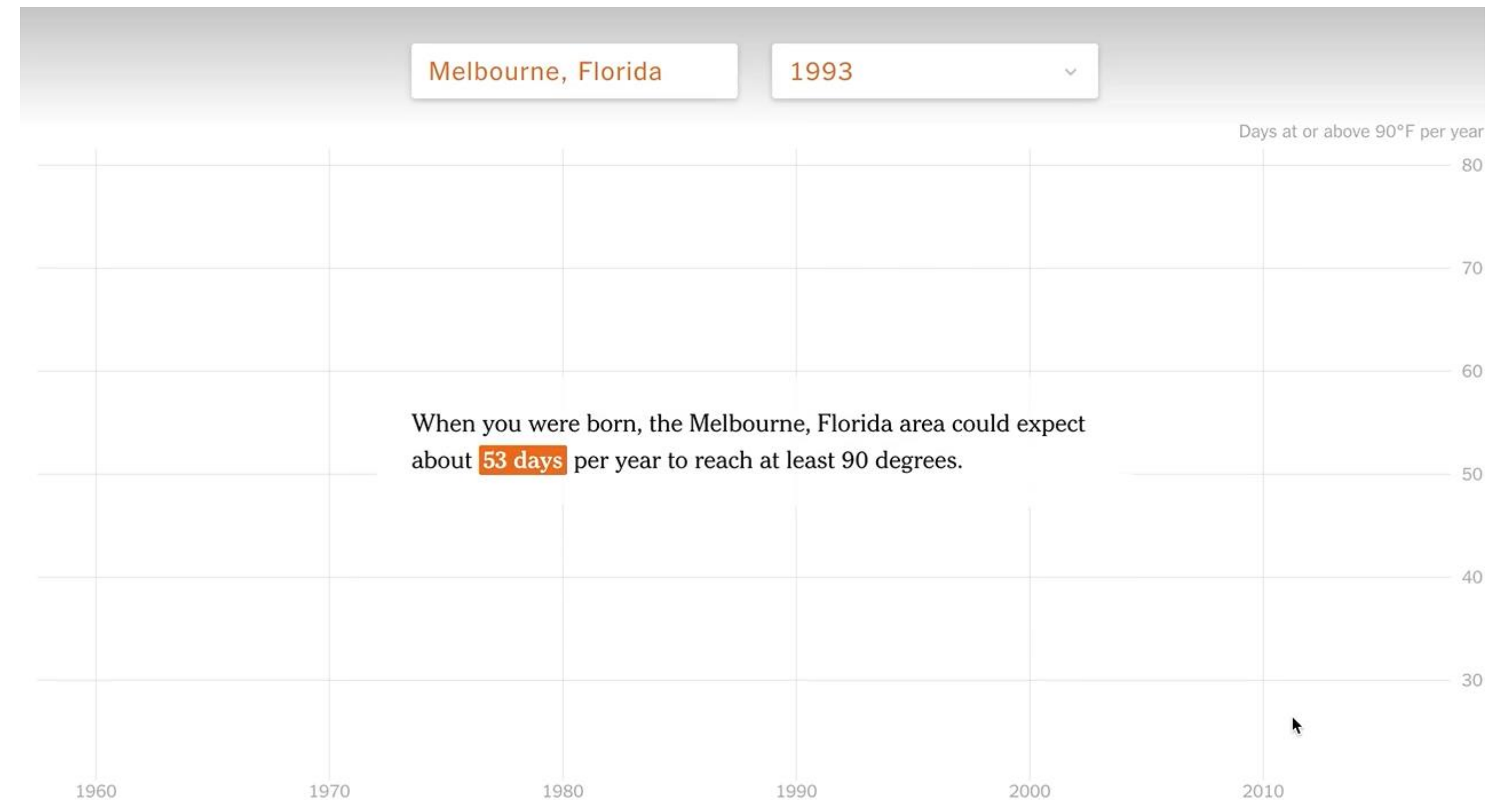
Run interactive simulations directly in the browser. No setup required.

Prompting Self-Reflection.

Help readers learn by asking them to reflect in a low pressure environment.

Reducing Cognitive Load.

Use effective representations to make complex topics more intuitive.



Popovich, Migliozi, Taylor, Williams, Watkins
NYTimes, 2018

Interactive Articles - Design

Five affordances of the format

Connecting People and Data.

Make data pleasant to work with. Happy readers are engaged readers.

Personalizing Reading.

Let readers choose the content that is relevant to their own experience.

Making Systems Playful.

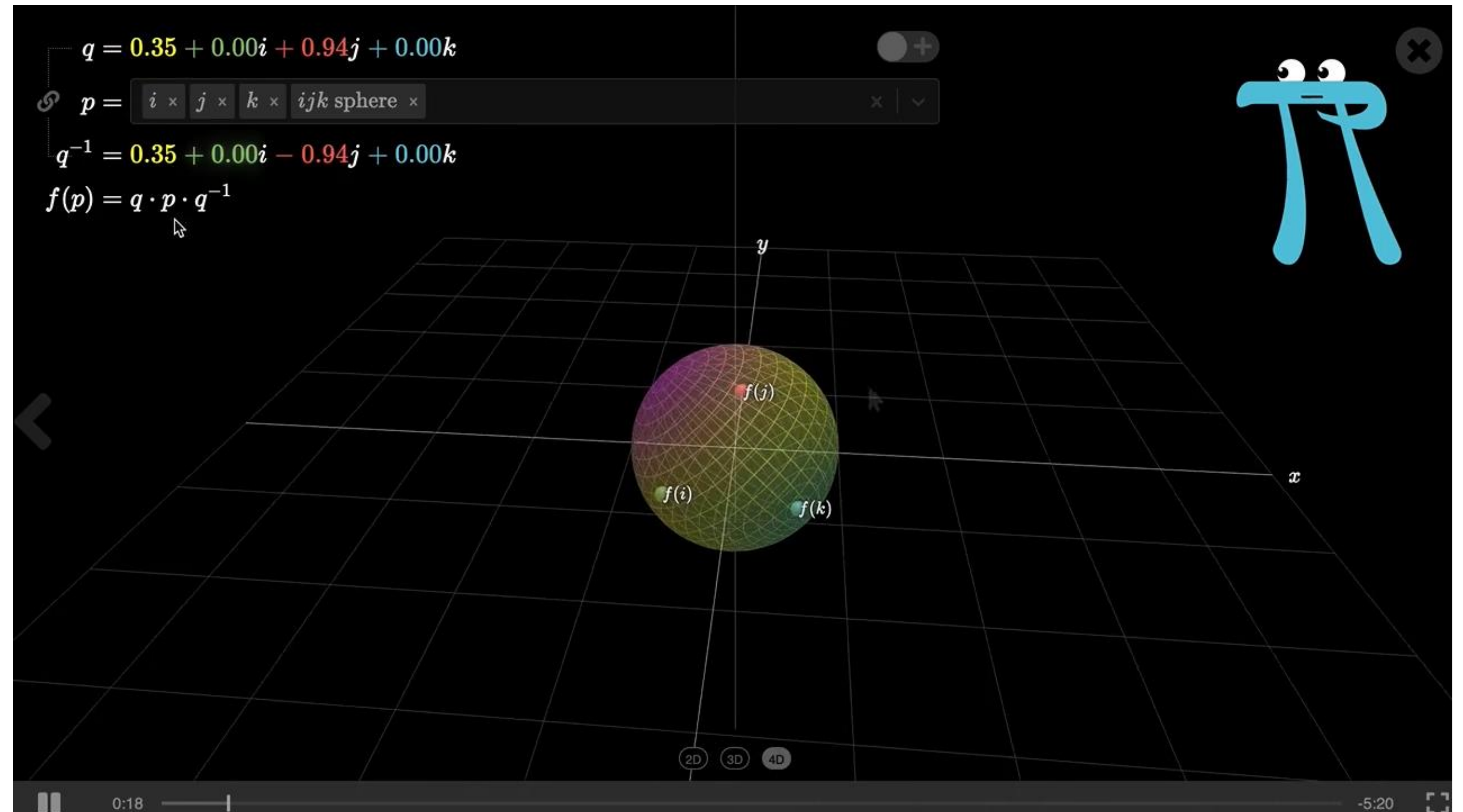
Run interactive simulations directly in the browser. No setup required.

Prompting Self-Reflection.

Help readers learn by asking them to reflect in a low pressure environment.

Reducing Cognitive Load.

Use effective representations to make complex topics more intuitive.



Interactive Articles - Design

Five affordances of the format

Connecting People and Data.

Make data pleasant to work with. Happy readers are engaged readers.

Personalizing Reading.

Let readers choose the content that is relevant to their own experience.

Making Systems Playful.

Run interactive simulations directly in the browser. No setup required.

Prompting Self-Reflection.

Help readers learn by asking them to reflect in a low pressure environment.

Reducing Cognitive Load.

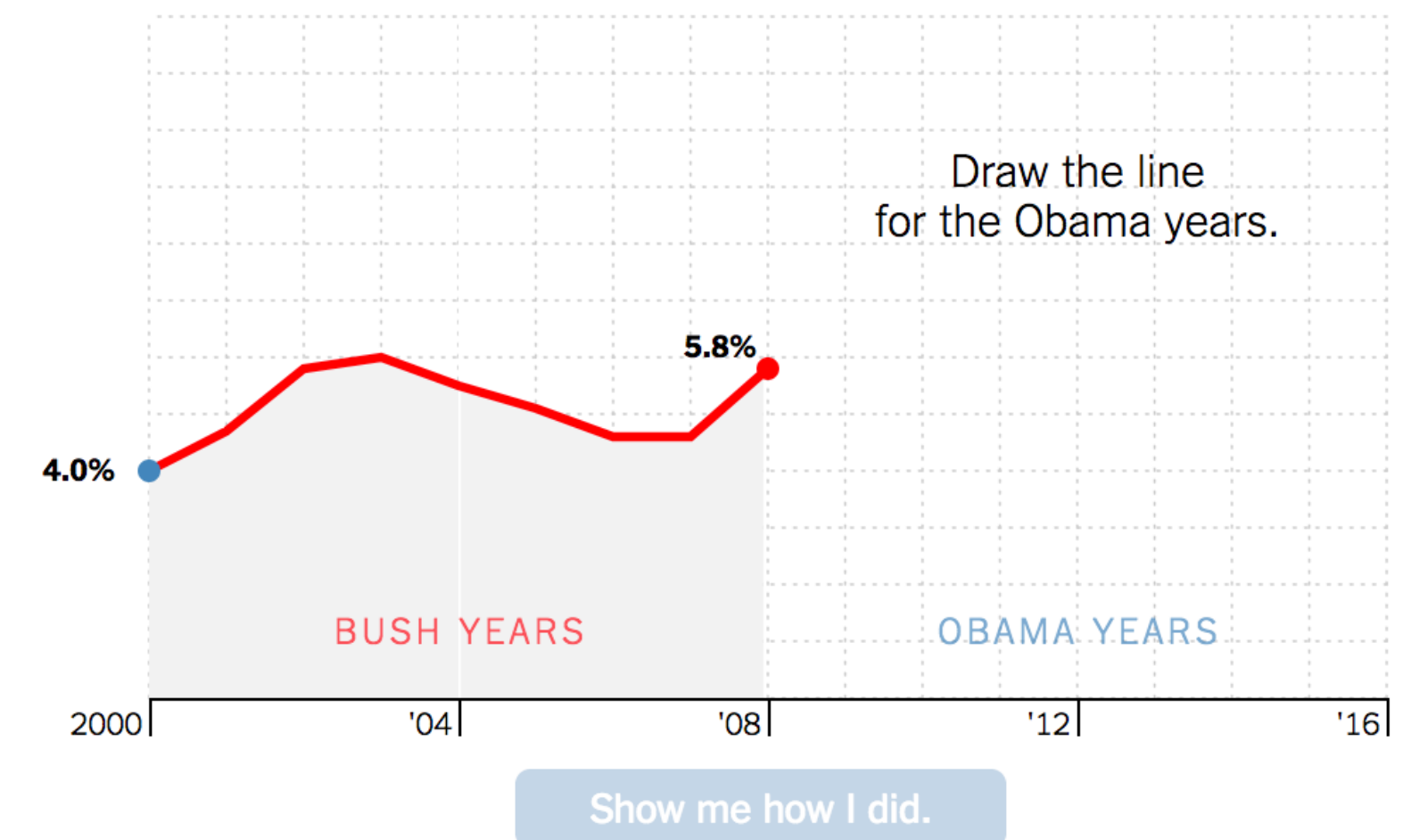
Use effective representations to make complex topics more intuitive.

You Draw It: What Got Better or Worse During Obama's Presidency

By LARRY BUCHANAN, HAEYOUN PARK and ADAM PEARCE JAN. 15, 2017

Draw your guesses on the charts below to see if you're as smart as you think you are.

Under President Obama, the **unemployment rate** ...



Interactive Articles - Design

Five affordances of the format

Connecting People and Data.

Make data pleasant to work with. Happy readers are engaged readers.

Personalizing Reading.

Let readers choose the content that is relevant to their own experience.

Making Systems Playful.

Run interactive simulations directly in the browser. No setup required.

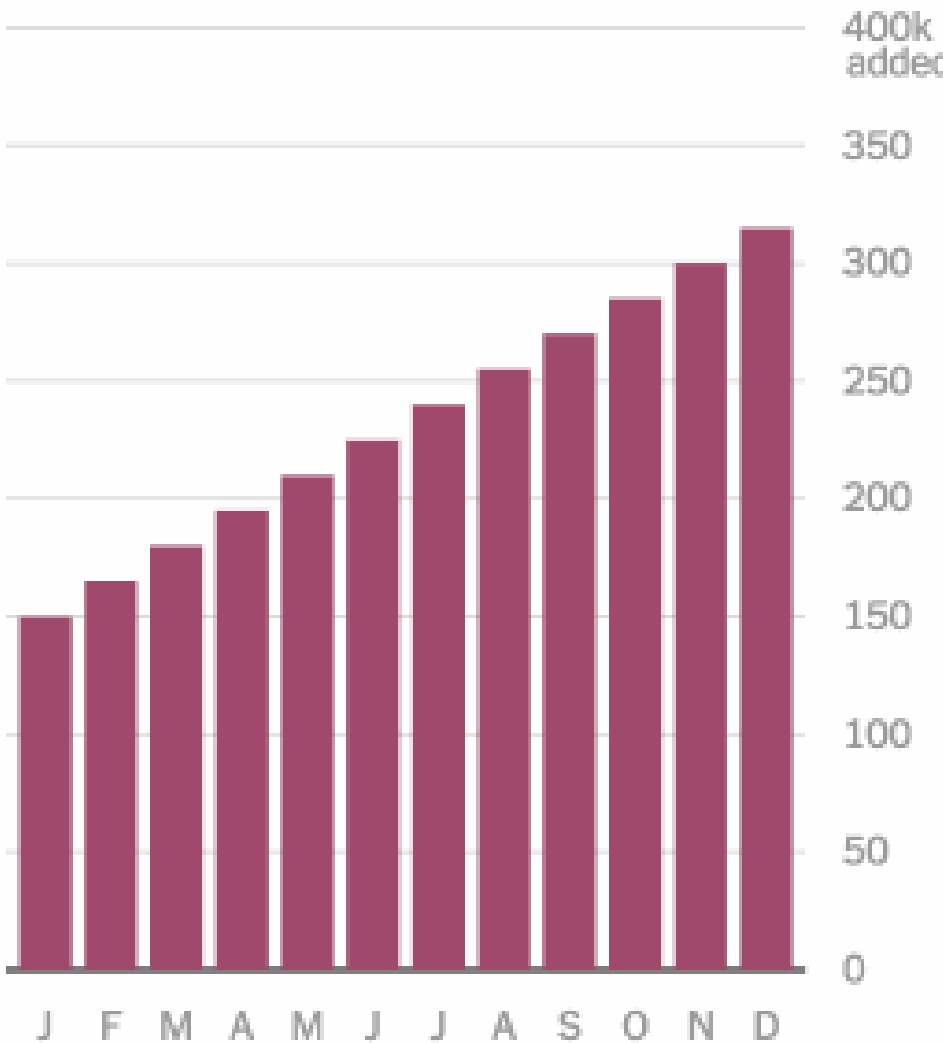
Prompting Self-Reflection.

Help readers learn by asking them to reflect in a low pressure environment.

Reducing Cognitive Load.

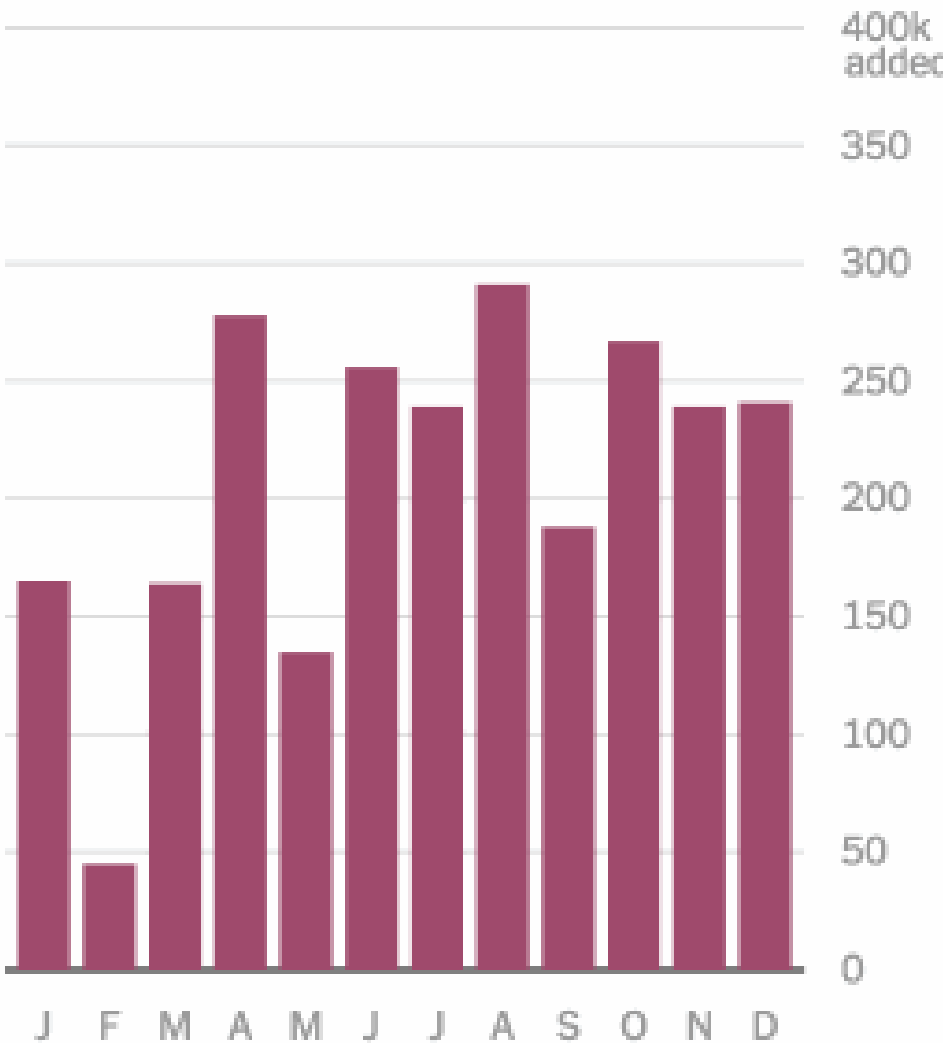
Use effective representations to make complex topics more intuitive.

If job growth **had been accelerating...**



...the jobs report could look like this:

Pause



Challenges

Few authoring tools available

Those that do exist are too low level or lack expressivity

Not suitable for production usage

Requires general purpose programming

In practice need to be a web developer

Difficult to evaluate

Limited empirical evidence; hard to analyze usage data; lack of metrics

Lack of design guidance

Integration between text and media

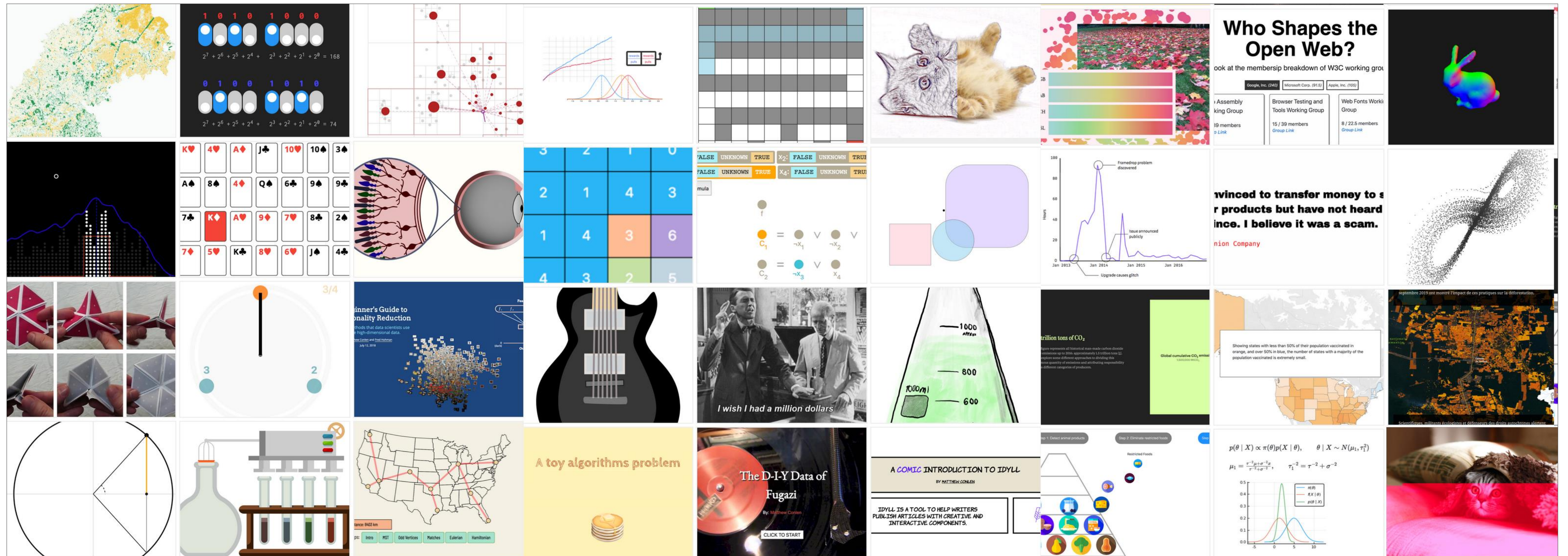
Multiple output targets

How to make content accessible?

Breadth of skills required (design, programming, narrative)

Idyll

A Markup Language for Authoring and Publishing Interactive Articles on the Web



DSL for Interactive Articles

Idyll Markup

```
[Header  
  title:“Hi Idyll!” /]
```

```
This is the text  
of my article.
```

```
[Graphic  
  value:x /]
```

```
And that was a  
custom graphic.
```

Input:

extended markdown

domain-specific graphics

DSL for Interactive Articles

Idyll Markup

```
[Header  
  title:“Hi Idyll!” /]
```

```
  
This is the text  
of my article.
```

```
  
[Graphic  
  value:x /]
```

```
  
And that was a  
custom graphic.
```

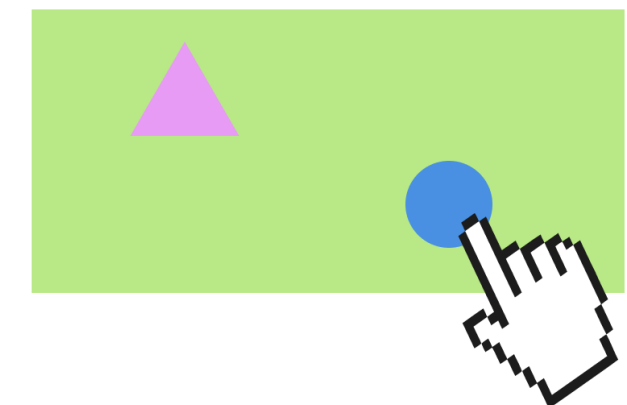
Input:
extended markdown
domain-specific graphics

Live Web Page

Hello Idyll

April 22, 2021

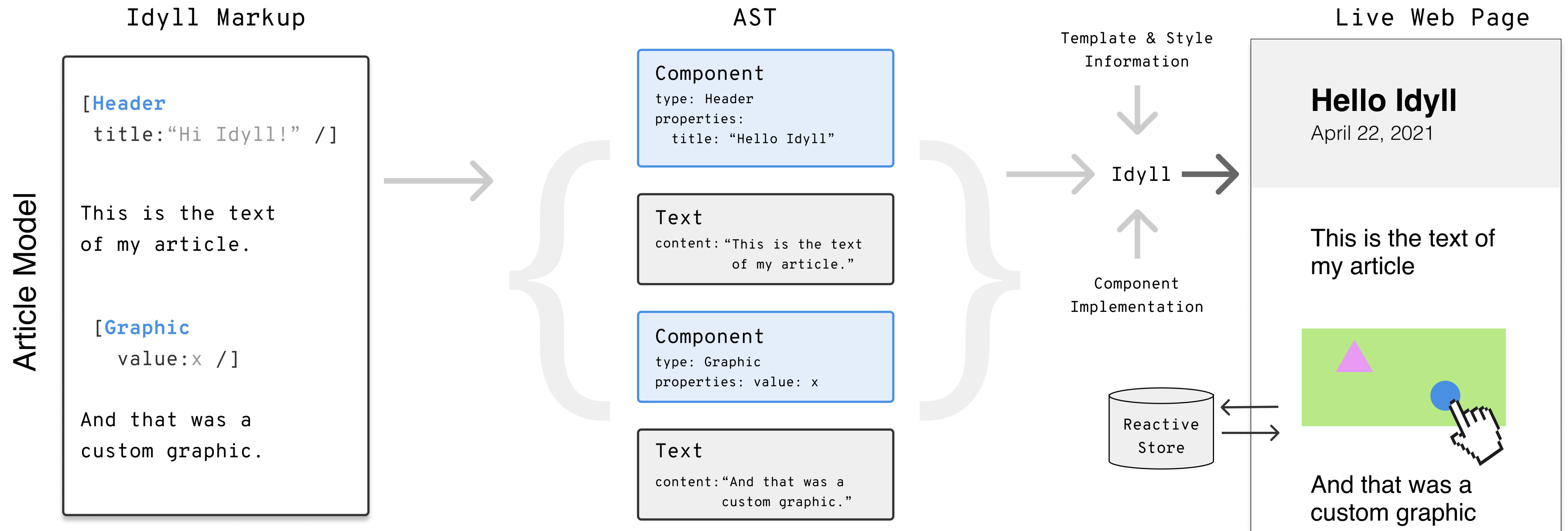
This is the text of
my article



And that was a
custom graphic

Output:
javascript,html,css

DSL for Interactive Articles



interactive document schema
(Interactions, linked text+graphics, events)

Input:
extended markdown
domain-specific graphics

Output:
javascript,html,css

Text + Parameterized Graphics

Write articles in
markdown.

Embed
parameterized
graphics.

Connect *reactive*
variables to user
input widgets to add
interactivity.

Text + Parameterized Graphics

Write articles in
markdown.

Embed *parameterized*
graphics.

Connect *reactive*
variables to user input
widgets to add
interactivity.

INPUT (EDITABLE)

```
# Hello World
```

```
[var name:"x" value:5 /]
```

```
The value of x is [Display value:x format:"d" /].
```

```
[Range value:x min:0 max:10 /]
```


Text + Parameterized Graphics

Write articles in
markdown.

Embed
parameterized
graphics.

Connect *reactive*
variables to user
input widgets to add
interactivity.

INPUT (EDITABLE)

```
# Hello World

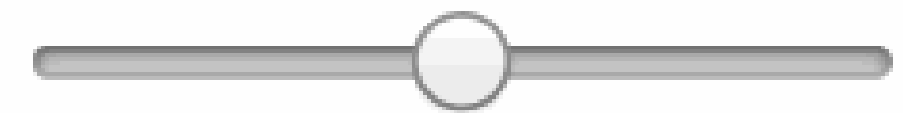
[var name:"x" value:5 /]

The value of x is [Display value:x format:"d" /].

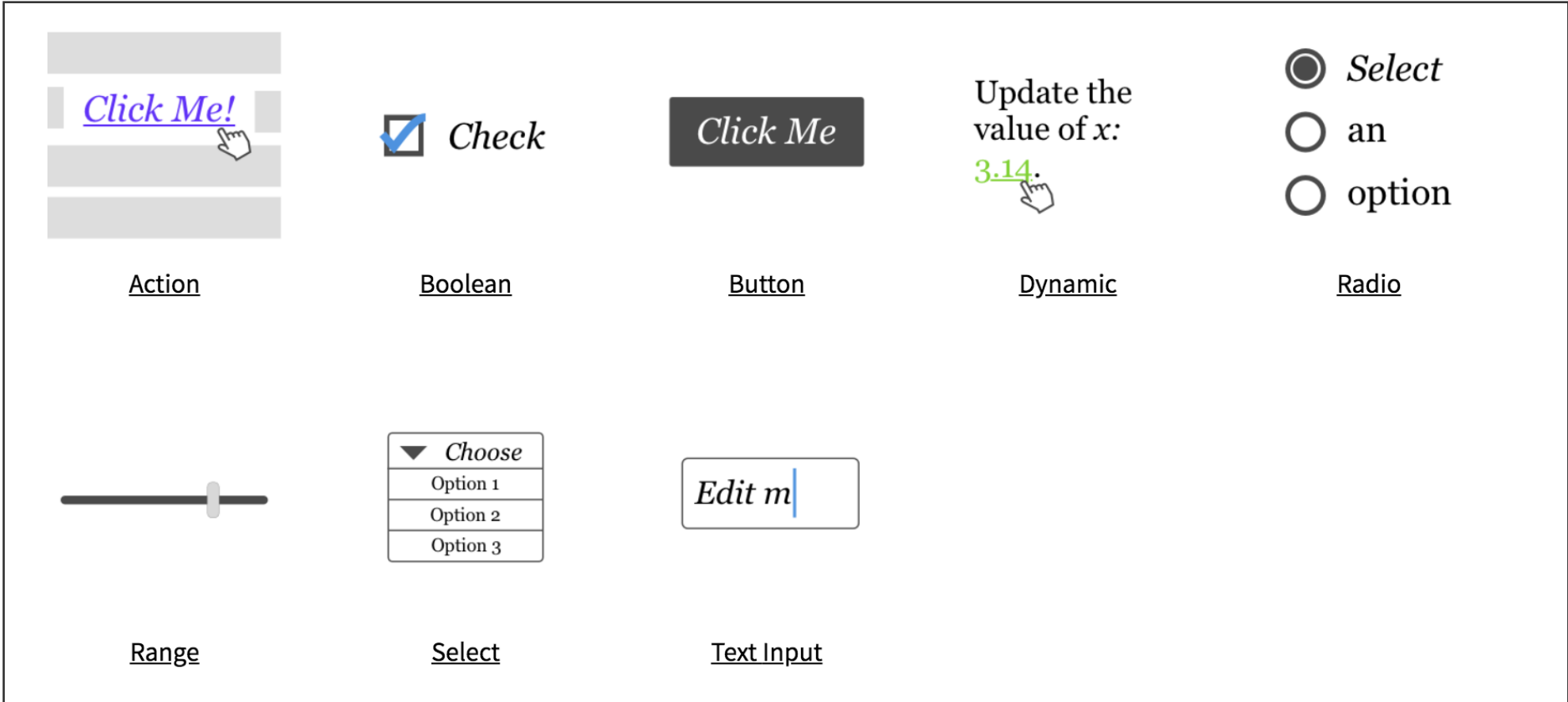
[Range value:x min:0 max:10 /]
```

Hello World

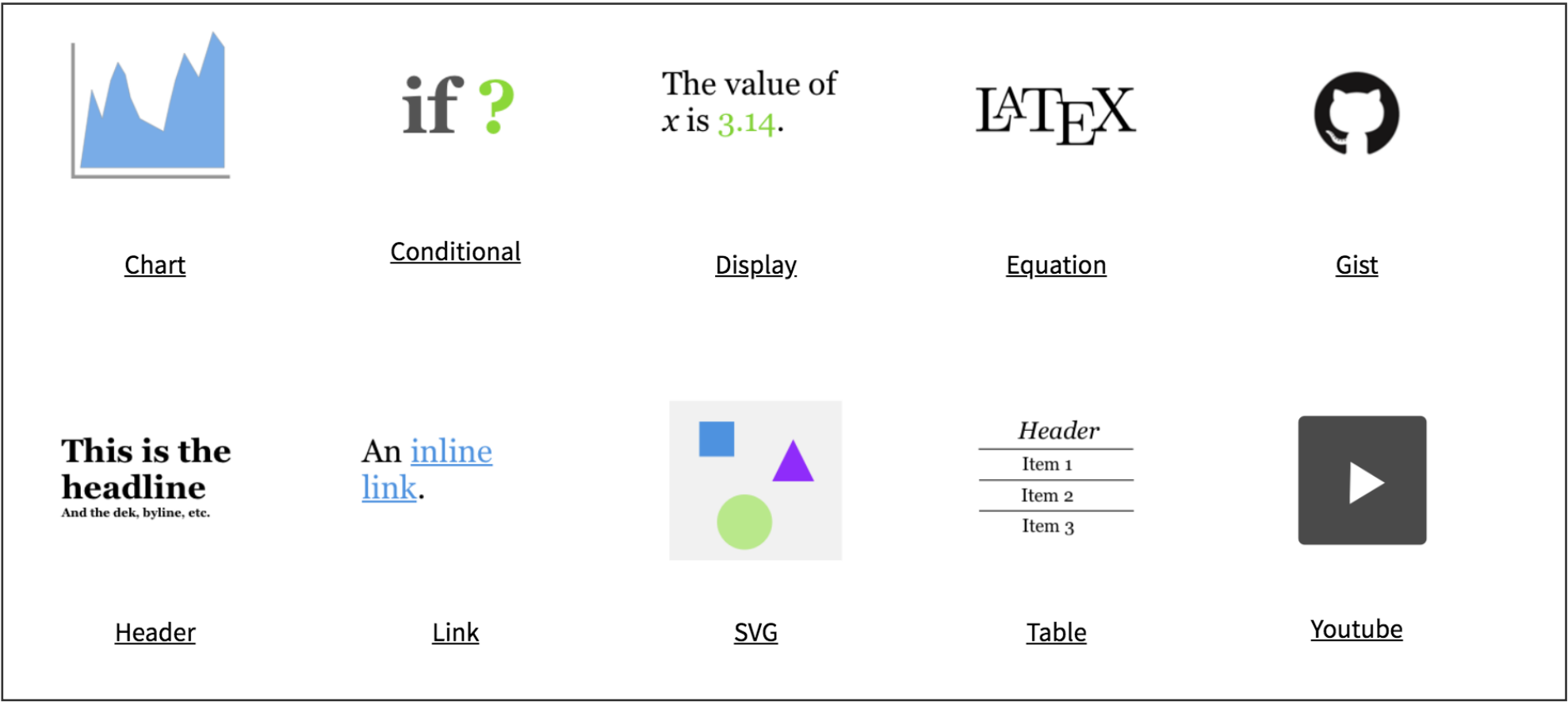
The value of x is 5.



Component Library



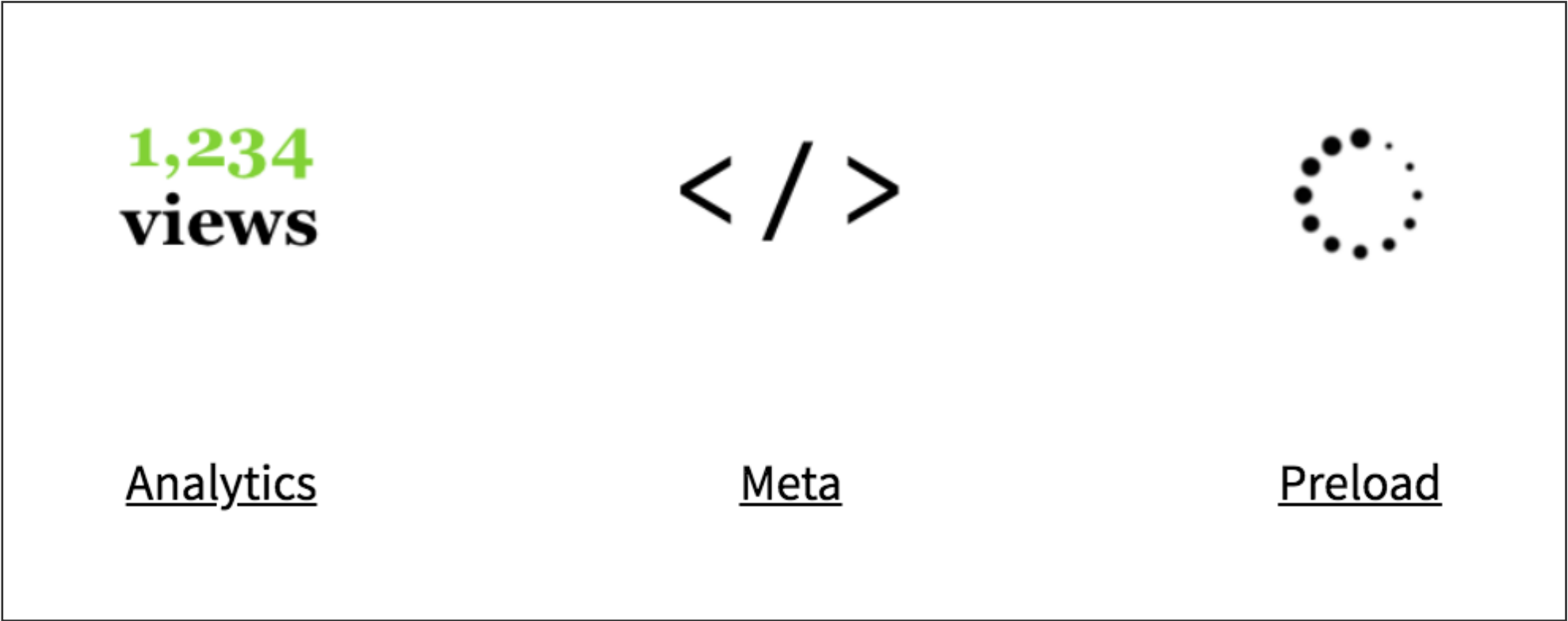
Input



Presentation



Layout



Helpers

Bring your own graphics

A JavaScript component API is exposed to end users.

Use with libraries like D3, Vega-Lite, Mapbox, Processing (P5), React, ...

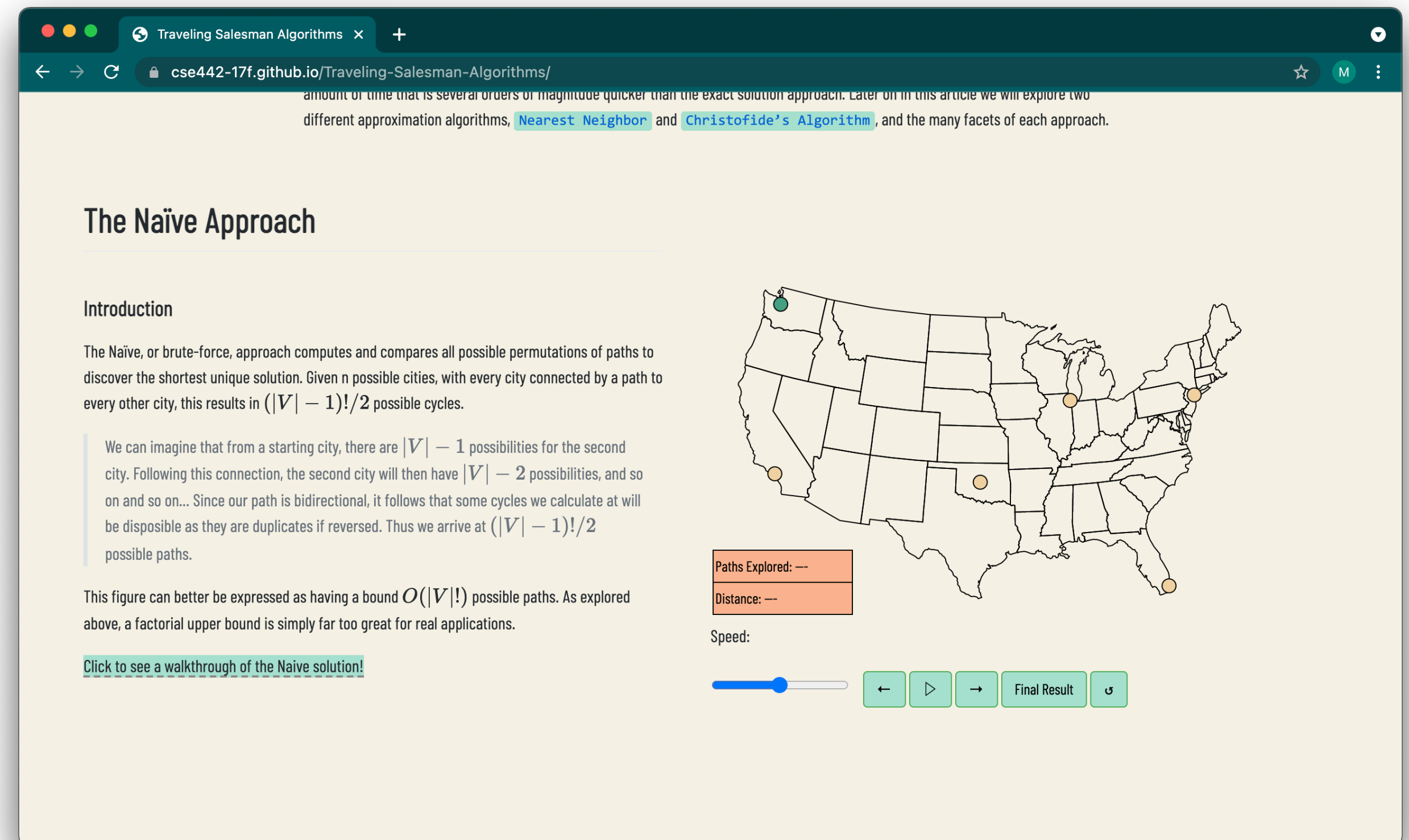


Usage

Refined design through
expert interviews.

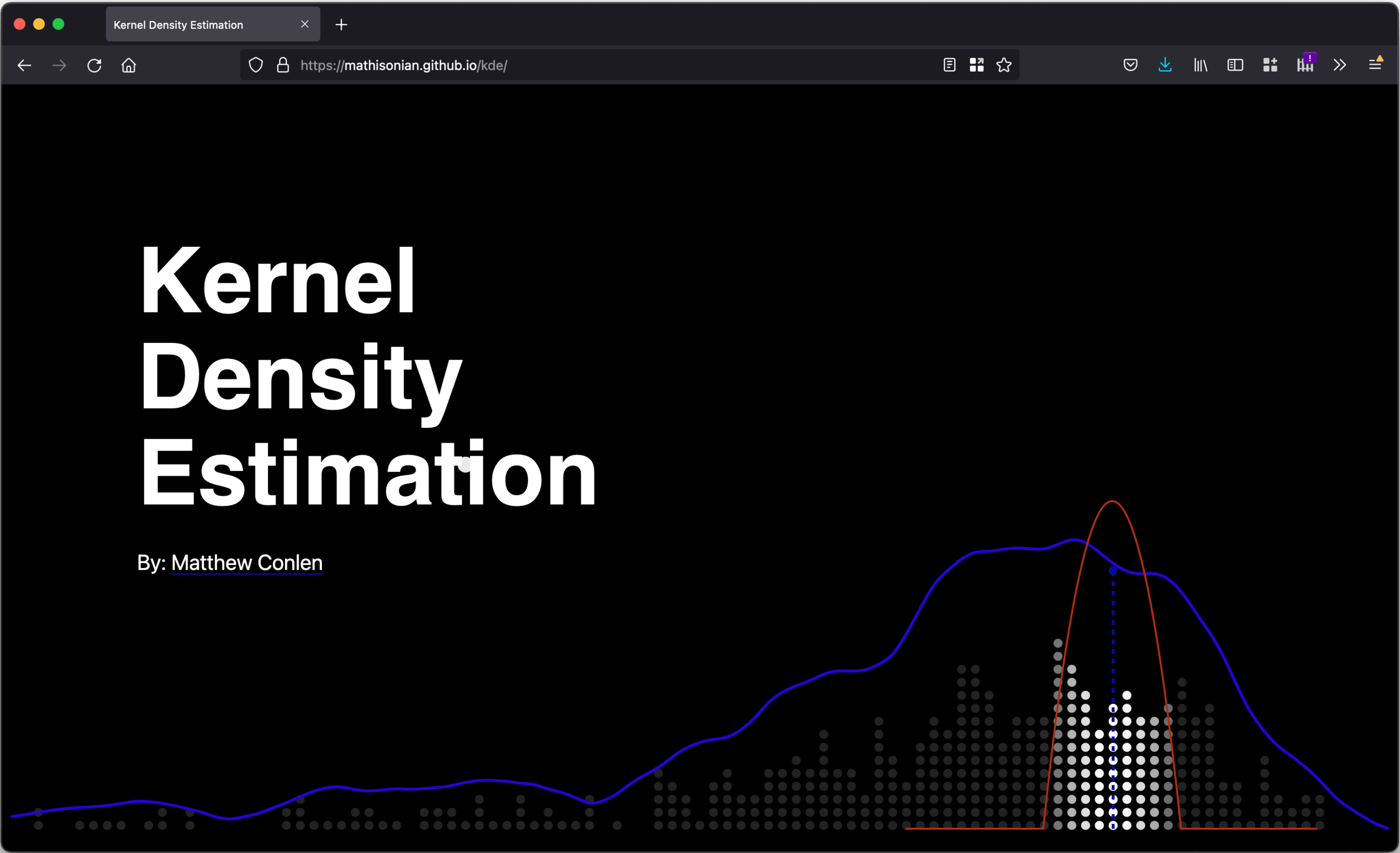
Deployed with undergrad CS
students at UW.

Students successfully in
creating highly customized
explorable explanations of
algorithms.



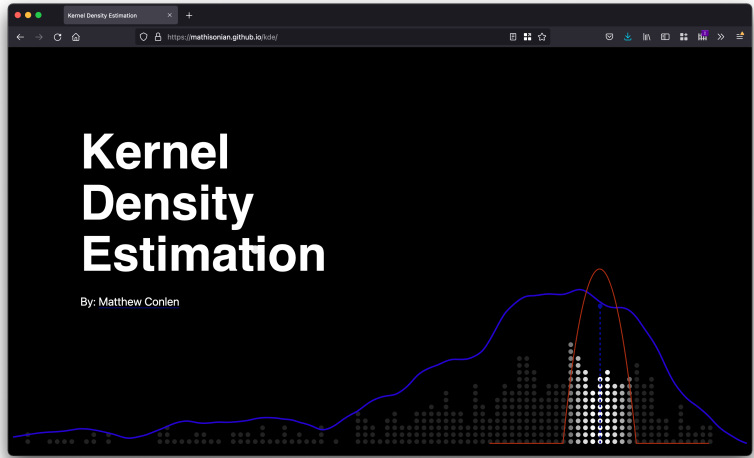
Mathew, Cherukupalli, Pusich, Zhao, *Traveling
Salesman Algorithms, CS442 '17*

Idyll in the wild



Kernel Density Estimation

Top search result under Wikipedia, many messages saying it helped people understand the topic.



Idyll...

Seems great but...

...still requires learning and writing markup syntax

...requires use of general-purpose programming tools, e.g. to make a new post, run development server

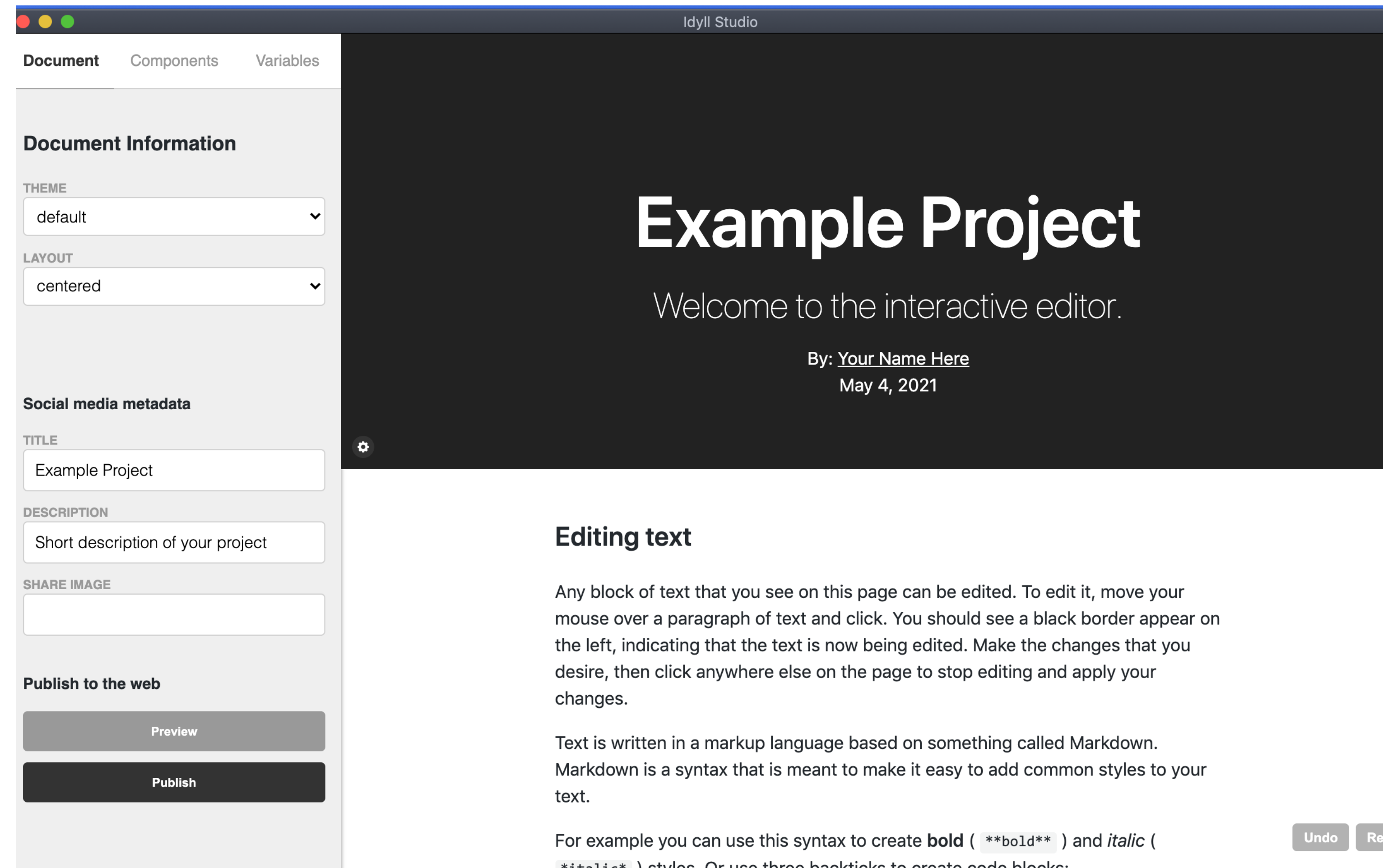
Idyll Studio

Built a **structured editor** for editing and creating Idyll programs

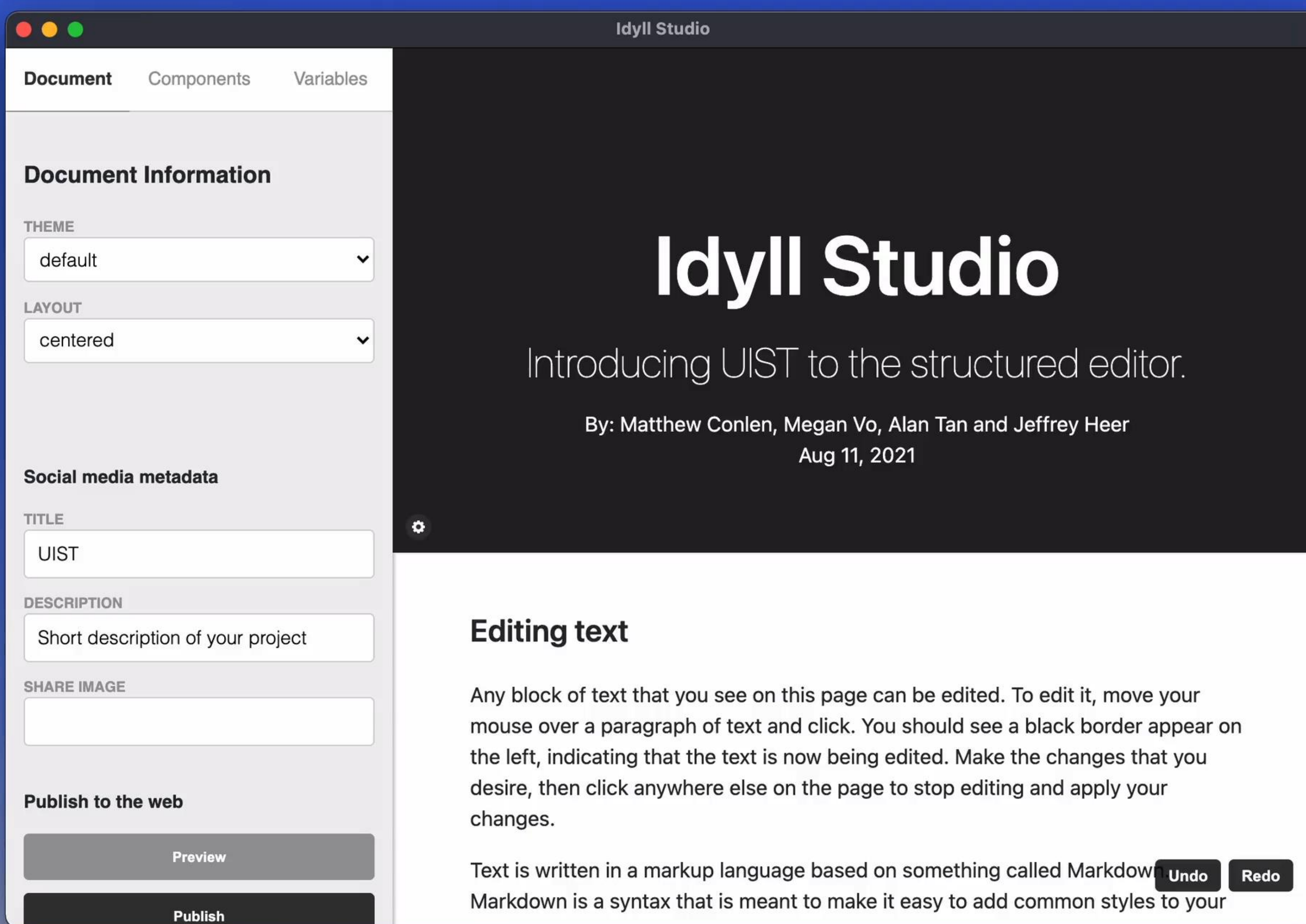
Reduce use of general-purpose programming tools.

Eliminate a class of syntax errors.

Reify Idyll model.



Conlen, Vo, Tan, Heer, *UIST 2021*



Idyll Studio Findings

1. The interface enabled users to **rapidly create designs with less stress**.

2. But some users struggled to synthesize multiple concepts. Others **needed time to understand the reactive model**.

3. The interface **should provide more visual feedback**, promote experimentation, provide documentation on demand, and have **guardrails to prevent likely mistakes**.

4. The interface empowered users and **provided better support to less technical users compared to existing tools**.

	Role	Tech.	I1	D1	I2	D2	I3	D3
P1	comp. biologist	4.2	y	y	y	y	y	y
P2	programmer	4.2	y	y	y	y	y	y
P3	hci researcher	3.8	y	y	y	y	y	y
P4	vis. practitioner	3.7	y	y	y	y	y	y
P5	hci researcher	3.5	y	y	y	y	y	y
P6	journalist	3.4	y	y	y	y	y	y
P7	hci researcher	3.4	y	y	y	y	y	y
P8	lab manager	3.4	y	y	y	y	y	y
P9	programmer	3.2	y	y	y	y	y	y
P10	vis. practitioner	2.9	y	y	y	y	y	y
P11	editor	2.9	y	y	y	y	y	y
P12	cs/journ. student	2.5	y	y	y	y	y	y
P13	data scientist	2.5	y	y	y	y	n	n
P14	analyst	2.3	y	y	y	y	n	n
P15	designer	2.3	y	y	y	y	n	n
P16	writer	1.8	y	y	y	n	n	n
P17	ux researcher	1.8	y	y	y	y	y	y
P18	designer	1.3	y	y	y	y	n	n

Assignment 3

Choose a topic to explain:

- ☐ *Dataset on a topic of interest*
- ☐ *Computational process*

Build and interactive article or explorable explanation:

- ☐ *A deliberate choice of narrative style use for exposition.*
- ☐ *Informal, consise writing designed for a broad audience.*
- ☐ *At least two distinct visualizations that update reactively based on user interactions.*
- ☐ *At least one use of animated transitions implemented in D3.*
- ☐ *At least one form of active user input.*

Deploy it online!

- ☐ *Svelte => Surge is the easiest way.*

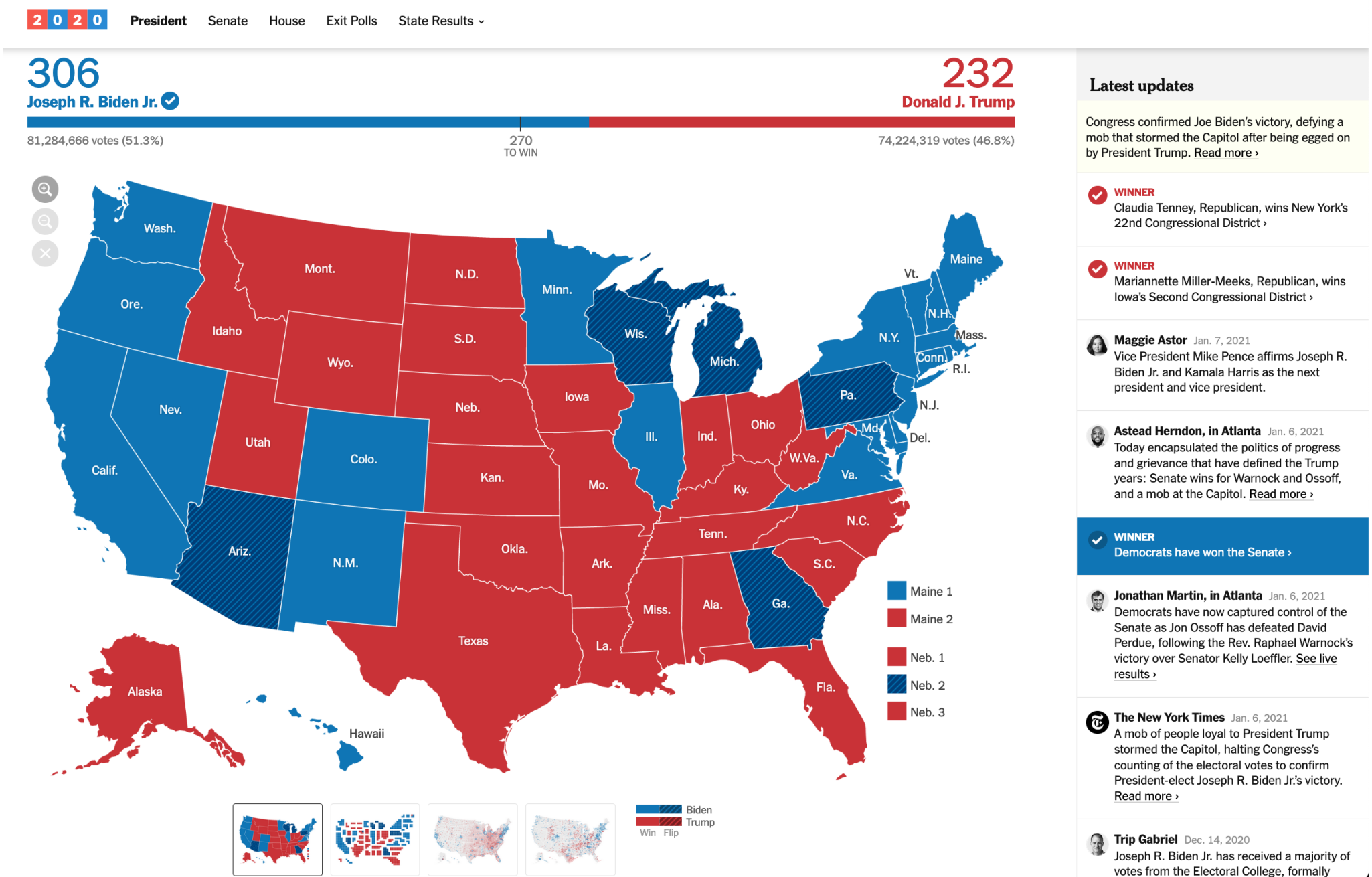
Design Prompt

Design Prompt ELECTION NIGHT

You want to keep the public as informed as possible of the status of the 2024 presidential election as it happens.

May want to convey:

- Current snapshot
- Likely outcomes
- Awareness of uncertainty
- Historical context
- Any unusual circumstances (e.g. delayed returns)



Design Prompt ELECTION NIGHT

Input Data

Vote counts

candidate, state, method, timestamp, # votes

Race calls

candidate, state, timestamp

Constraints

Near real time

Never seen the data before

Data may contain errors

Should not erode public trust

Design Prompt ELECTION NIGHT

Input Data

Vote counts

candidate, state, method, timestamp, # votes

Race calls

candidate, state, timestamp

Constraints

Near real time

Never seen the data before

Data may contain errors

Should not erode public trust

**How would you
visualize this?**

**What data would you
use?**

Design Prompt ELECTION NIGHT

Input Data

Vote counts

candidate, state, method, timestamp, # votes

Race calls

candidate, state, timestamp

Constraints

Near real time

Never seen the data before

Data may contain errors

Should not erode public trust

**How would you
visualize this?**

**What data would you
use?**

You want to keep the public as informed as possible of the status of the 2024 presidential election as it happens.

May want to convey:

Current snapshot

Likely outcomes

Awareness of uncertainty

Historical context

Any unusual circumstances